Wireless Transmission:

Physical Layer Aspects and Radio Channel Characteristics

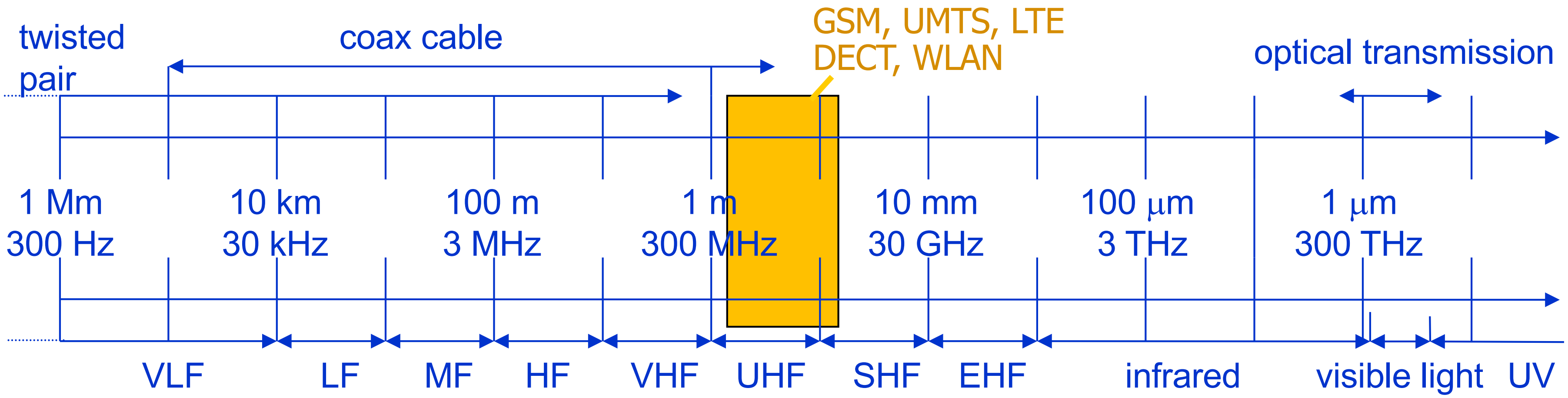
- Frequencies and Signals
- Signal Propagation
- Antennas
- Channel Capacity – Signal vs. Interference
- Multiplexing
- Modulation and Coding
- Spectrum Spreading
- Cellular Networks

Radio Frequencies and Signals

Questions to be answered:

1. What are the wavelengths of the different band used by GSM and WLAN?
2. Which bands typically provide better penetration of walls?
3. What´s the disadvantage of using higher frequencies for communications? What´s the advantage?
4. Sketch the frequency spectrum of a 100 MHz sinus signal and of a symmetric rectangular signal with the same period.

Frequencies for communication



VLF = Very Low Frequency
LF = Low Frequency
MF = Medium Frequency
HF = High Frequency
VHF = Very High Frequency

UHF = Ultra High Frequency
SHF = Super High Frequency
EHF = Extra High Frequency
UV = Ultraviolet Light

Relation between frequency f and wave length λ:

$\lambda = c / f$, with speed of light $c \cong 3 \times 10^8$ m/s

Important: the wave length highly influences the size of antennas!

Frequencies for mobile communication

VHF/UHF-ranges for mobile radio (<3GHz)

- simple & small antennas
- good propagation characteristics (limited reflections, small path loss, good penetration of walls)

SHF and higher (>3 GHz, mmWaves) for directed radio links, satellite communication

- small antenna, strong focus
- larger bandwidth available
- no penetration of walls

Systems for EHF (beyond 30 GHz)

- limitations due to absorption by water and oxygen molecules (resonance frequencies) resulting in weather dependent fading, signal loss caused by heavy rainfall etc.

Signals in general

- physical representation of data
- function of time and location
- signal parameters: parameters representing the value of data
- classification
  - continuous time vs. discrete time
  - continuous values vs. discrete values
  - analog signal = continuous time and continuous values
  - digital signal = discrete time and discrete values
- signal parameters of periodic signals:
  period T, frequency f=1/T, amplitude A, phase shift φ
  - sine wave as special periodic signal for a carrier:

$$s(t) = A_t \sin(2 \pi f_t t + \varphi_t)$$

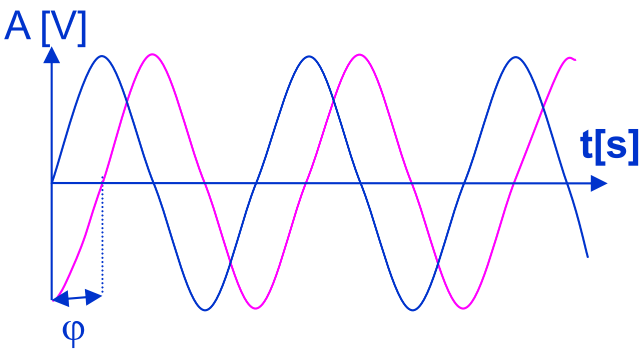
amplitude

frequency

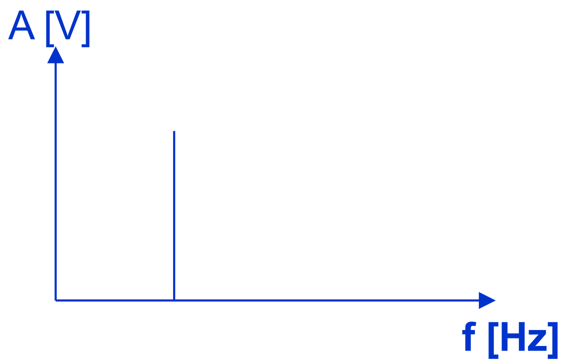
phase shift

Different representations of signals

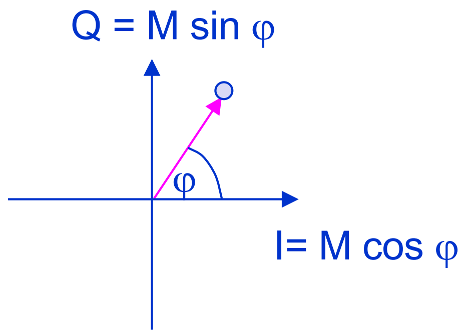
Time domain
(amplitude over time)



Frequency/spectrum domain



Phase diagram
(amplitude M and phase φ in polar coordinates)



Time domain:
- Shows two periodic signals with same frequency but different phase
- Can be used to display any signal (periodic or not) over time

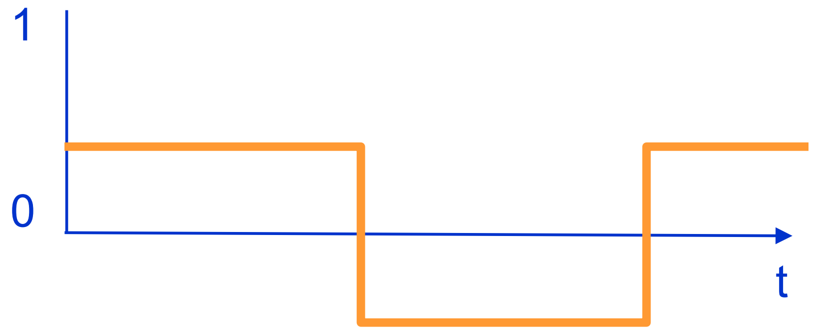
Frequency/spectrum domain:
- Shows the frequency of a periodic signal (single frequency)
- We will use this later to show the frequency range (spectrum) used by frequency modulated signals

Phase diagram:
- Shows the phase of a signal within the 360° phase range
- We will use this later to show the phases used by phase modulated signals

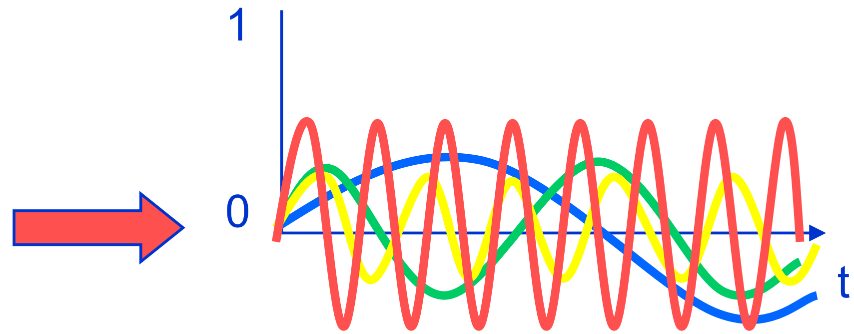
Fourier representation of periodic signals

Every periodic signal g(t) can be constructed by

$$g(t) = \frac{1}{2}c + \sum_{n=1}^{\infty} a_n \sin(2\pi nft) + \sum_{n=1}^{\infty} b_n \cos(2\pi nft)$$



ideal periodic signal



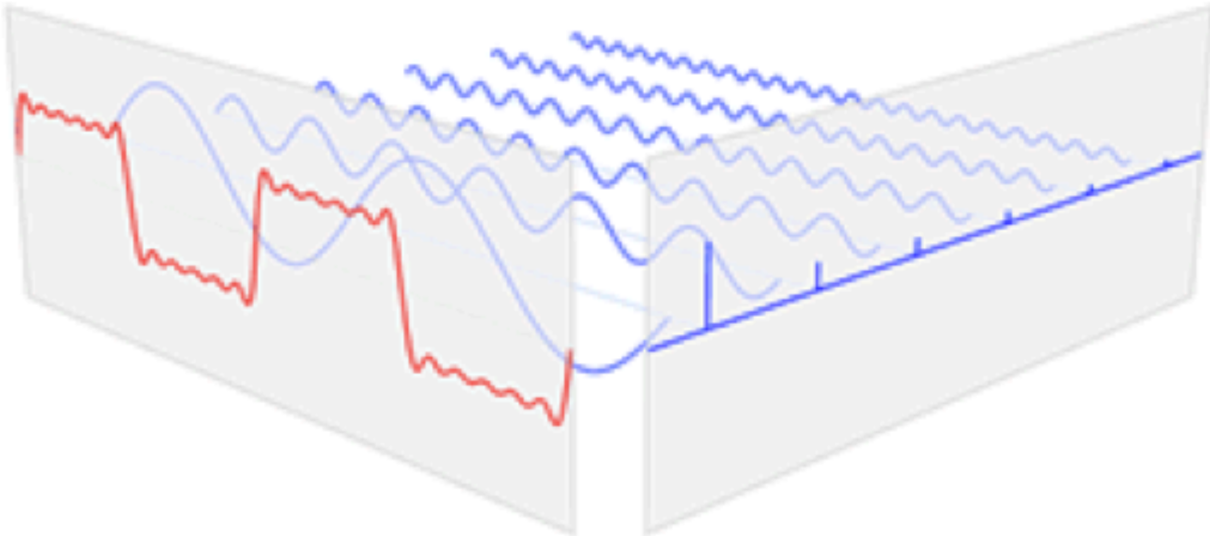
real composition
(based on harmonics)

- Composed signals are transferred into frequency domain using Fourier transformation
- Digital signals need
  - infinite frequencies for perfect transmission
  - modulation with a carrier frequency for transmission (analog signal!)

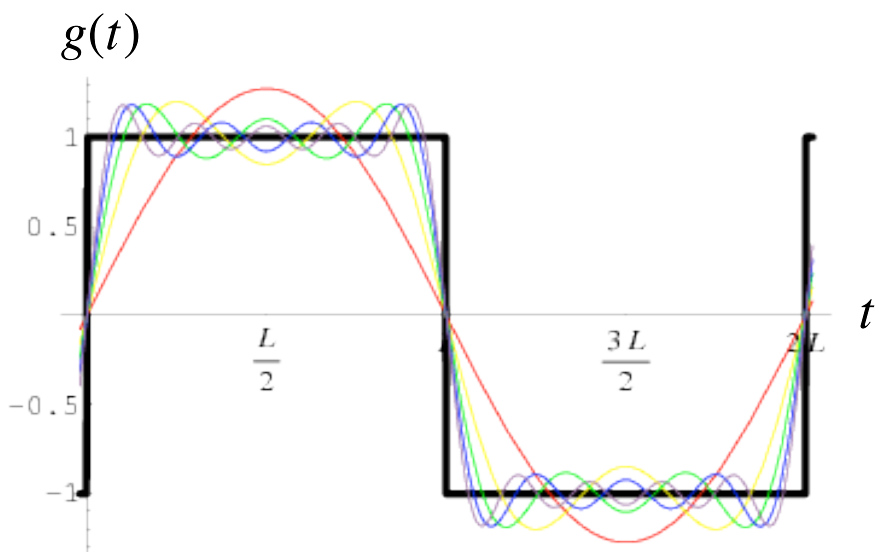
Fourier representation of periodic signals

Example

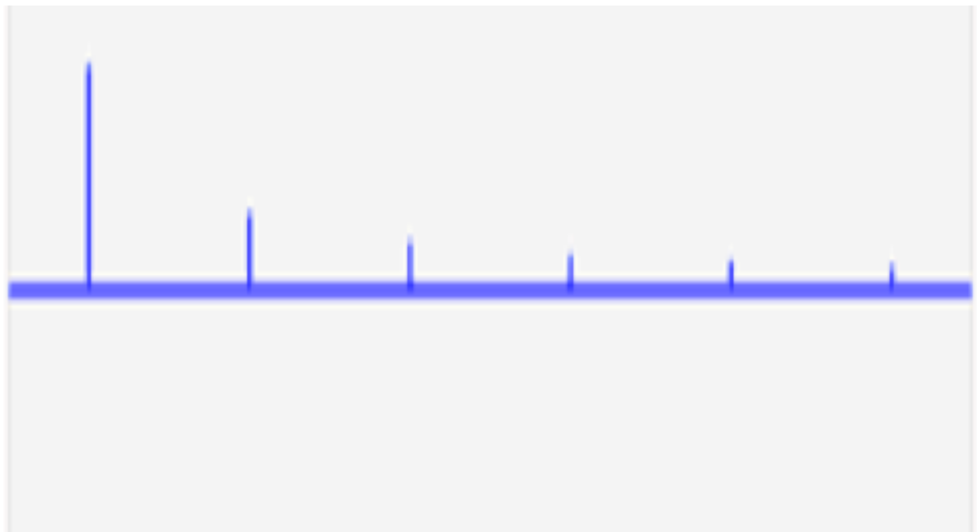
$$g(t) = \frac{4}{\pi} \sum_{n=1,3,5,...}^{\infty} \frac{1}{n} \sin(\frac{\pi n t}{L})$$



Time domain



Frequency domain

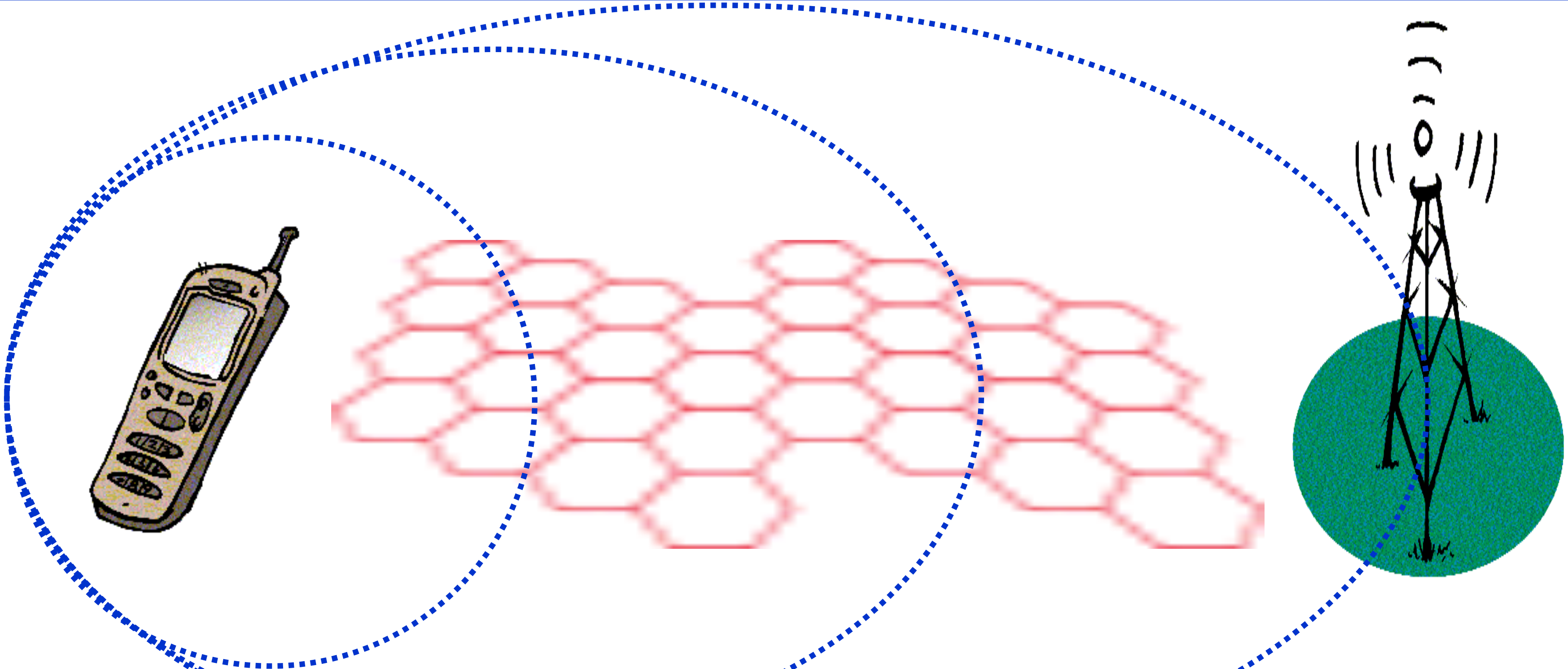


Signal propagation

Questions to be answered:

1. Why is a pathloss proportional to d^2 is assumed for line-of-sight connections? Why is d^2 considered as the best case?
2. What are the reasons for fast fading? What is exactly happening?
3. Why does fast fading depend on mobility? Why is fast fading frequency-dependent?
4. What is the difference between fading due to simple phase shifts between arriving signals and inter symbol interference?
5. What are the factors of the received signal important for error-free reception?

Signal propagation: received power due to pathloss

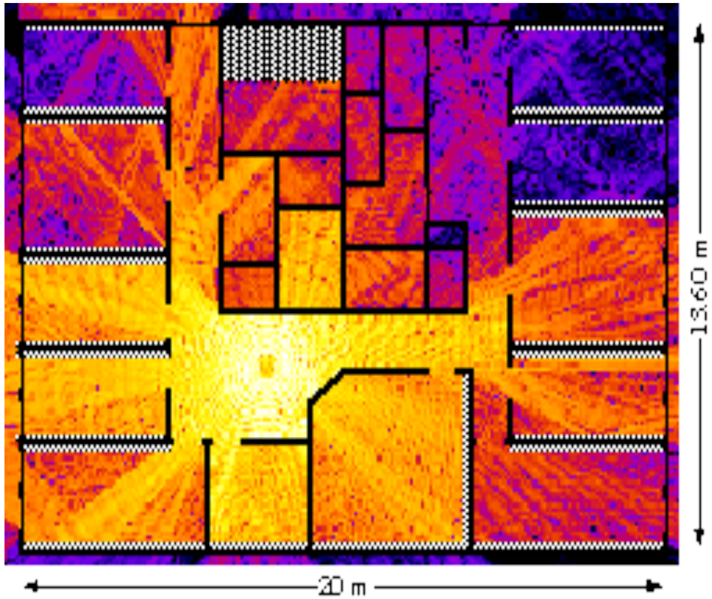


	1m	10m	100m
Ideal line-of-sight (d^{-2}):	1	1:100	1:10000
Realistic propagation ($d^{-3.5 \ldots 4}$):	1	1:3000 to 1:10000	1:10 Mio to 1:100 Mio

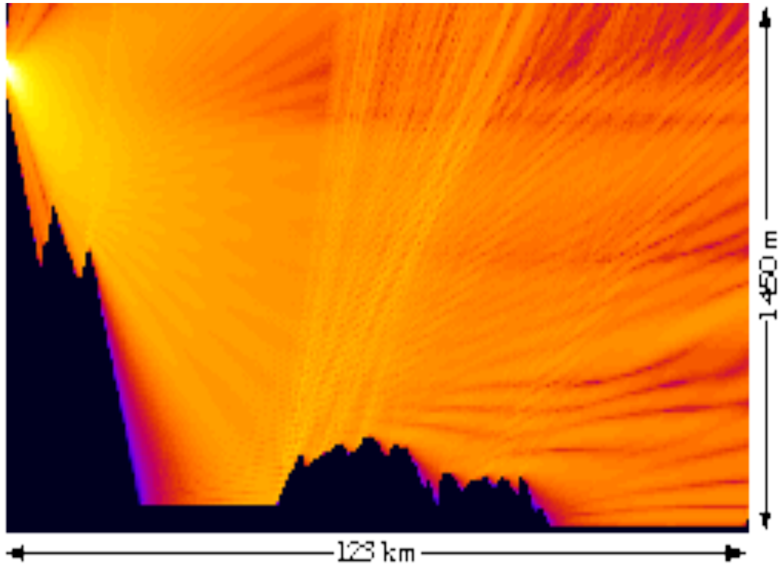
35-40 dB (1m ↔ 10m) 35-40 dB (10m ↔ 100m)

Note: the pathloss applies to the interference as well!

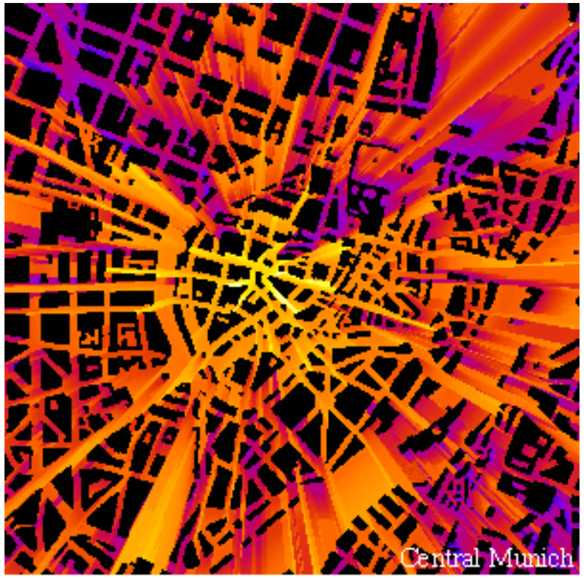
Real world examples with reflections



2D indoor



3D outdoor (TV sender on top of a mountain



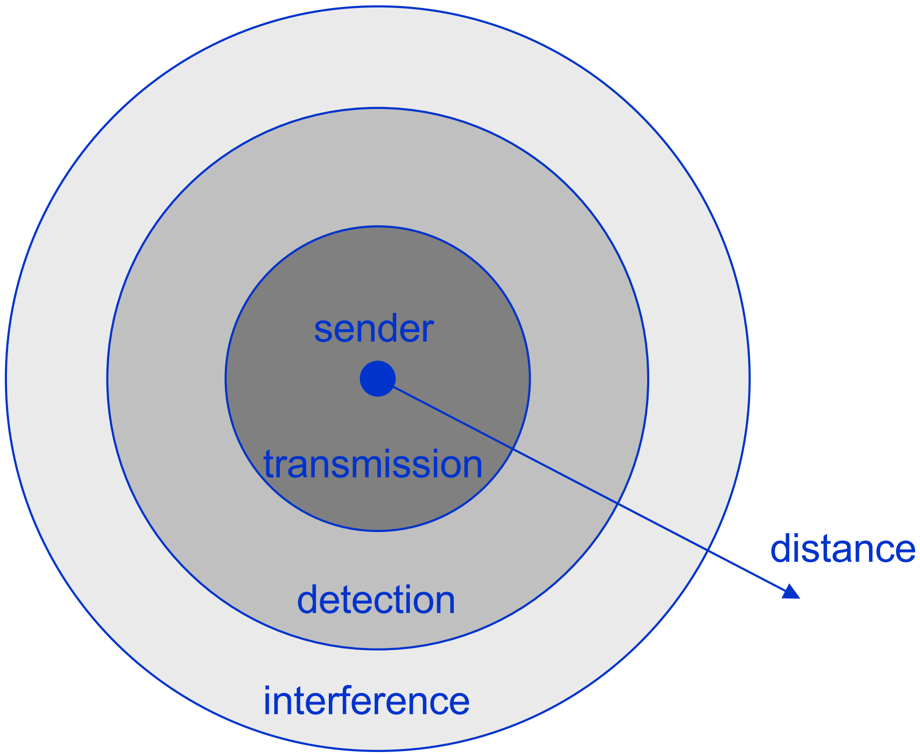
2D outdoor (inner city scenario)

Colors represent different signal strength

- bright yellow represents best
- dark blue represents weakest signal

Signal propagation ranges

- Transmission range
  - communication is possible
  - low error rate
- Detection range
  - detection of the signal as such is possible
  - no communication possible
- Interference range
  - signal may not be detected
  - signal just adds to the background noise



There are two requirements for a successful transmission:

- the signal to interference & noise ratio SINR is above a threshold (this depends on the radio technology (modulation, coding), HW and signal processing capabilities of the receiver)
- the received signal strength S is above a threshold (this is defined by the sensitivity of the receiver)

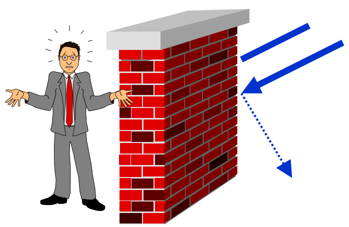
Signal propagation for the static case (no mobility)

Propagation in free space is similar to visible light (straight line)

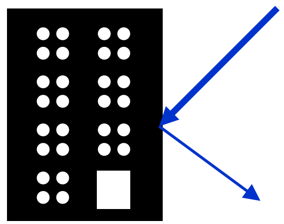
Receiving power is proportional to $1/d^{2 \ldots 4}$
(where d = distance between sender and receiver)

Received power is additionally influenced by

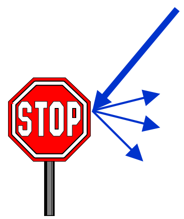
- shadowing
- reflection at large obstacles
- scattering at small obstacles
- diffraction at edges



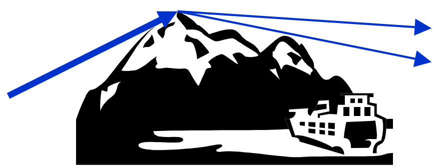
shadowing



reflection



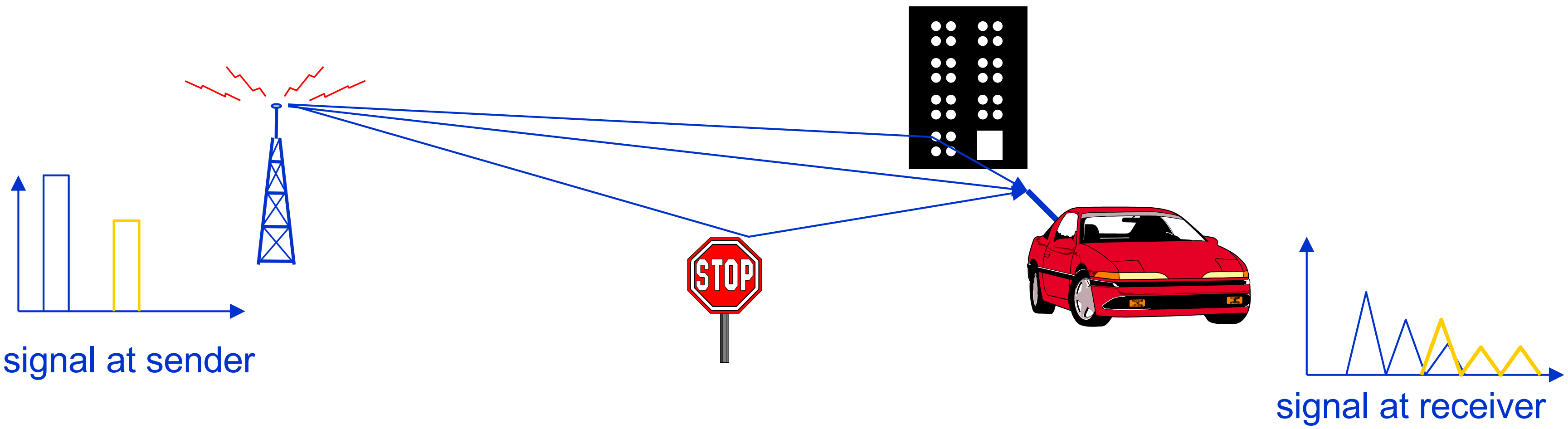
scattering



diffraction

Multipath propagation (static case)

Due to reflection, scattering and diffraction, the signal may take many different paths – and with different distances – between sender and receiver



Time dispersion over symbols: the signal is dispersed over parts of the symbol

➔ interference with "neighbor" symbols, Inter Symbol Interference (ISI) depends on the degree of shift between symbols

The signal reaches a receiver directly and phase shifted, e.g. 0 and 180° phase shift

➔ distortion depends on the phases of the different multipath parts

Effects of mobility – Fast and slow fading

Due to mobility, channel characteristics change over time and location

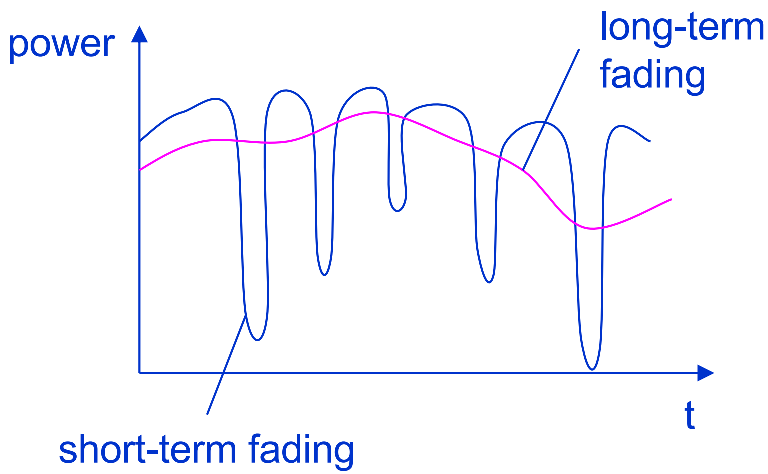
- signal paths change
- different delay variations of different signal parts (frequencies)
- different phases of signal parts

➔ results in quick changes in the power received (called short-term fading or fast fading)

Additional changes in

- distance to sender
- obstacles further away

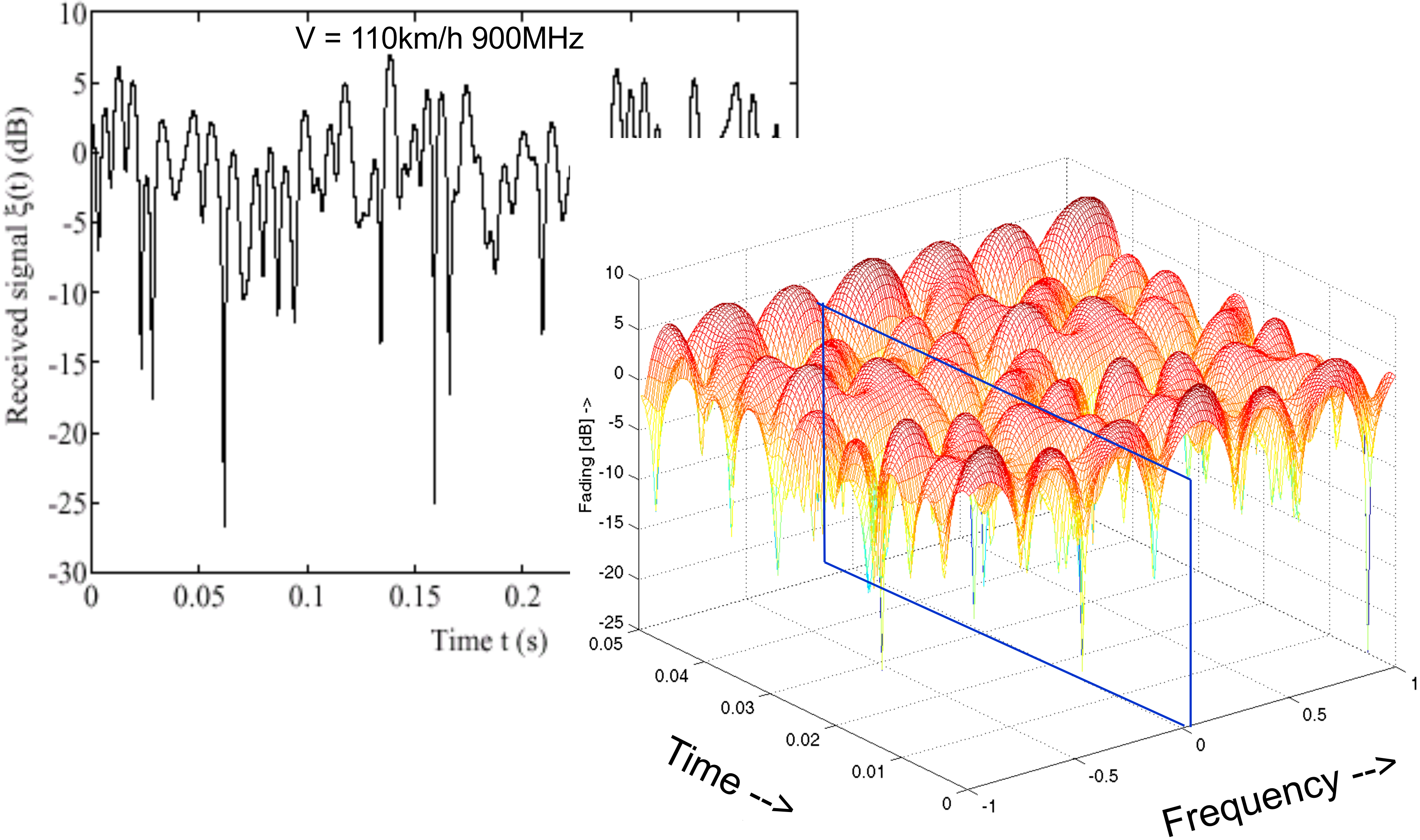
➔ results in slow changes in the average power received (called long-term fading or slow fading)



Time and frequency dependency of fast fading

Fast fading is time- and frequency-dependent!

(examples show Rayleigh fading for urban environments)



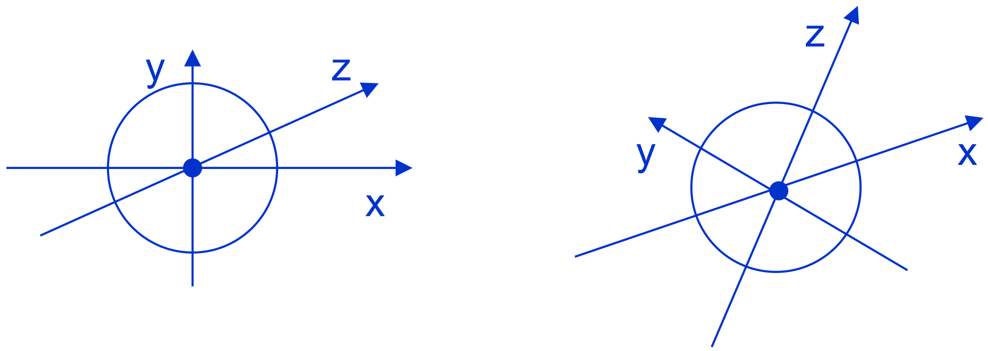
Antennas

Questions to be answered:

1. What are the characteristics of an isotropic radiator?
2. What directional effects show simple practical antennas?
3. How can you build a directional antenna from 2 simple $\lambda/2$ antennas? What are the prerequisites?

Antennas: isotropic radiator

- Radiation and reception of electromagnetic waves, coupling of wires to space for radio transmission
- Isotropic radiator: equal radiation in all directions (three dimensional) - only a theoretical reference antenna
- Real antennas always have directive effects (vertically and/or horizontally)
- Radiation pattern: measurement of radiation around an antenna



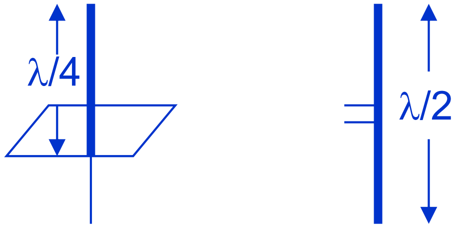
ideal
isotropic
radiator

Antennas: simple dipoles

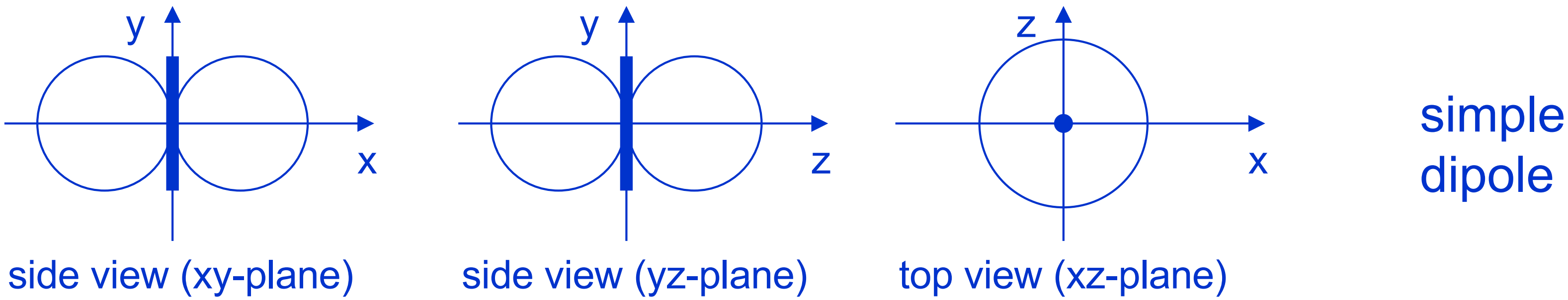
Real antennas are not isotropic radiators but, e.g.

- dipoles with lengths $\lambda/4$ on car roofs or
- $\lambda/2$ as Hertzian dipole

➔ shape of antenna proportional to wavelength



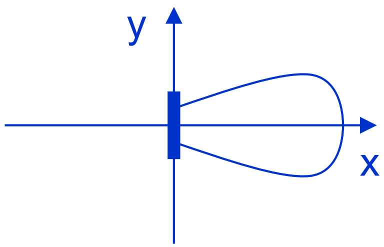
Example: Radiation pattern of a simple Hertzian dipole



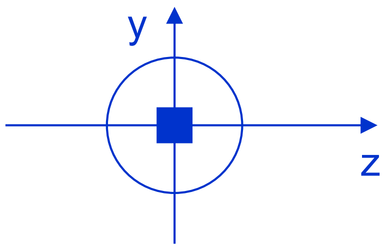
Antenna gain: maximum power in the direction of the main lobe compared to the power of an isotropic radiator (with the same average power)

Antennas: directed and sectorized

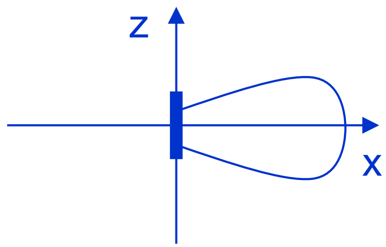
Often used for microwave connections or base stations for mobile phones (e.g. radio coverage of a valley)



side view (xy-plane)

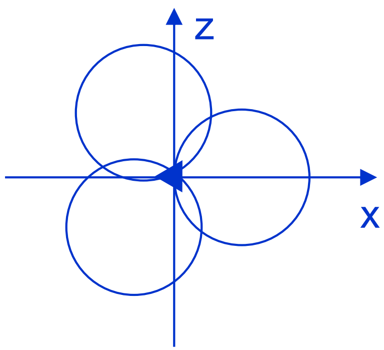


side view (yz-plane)

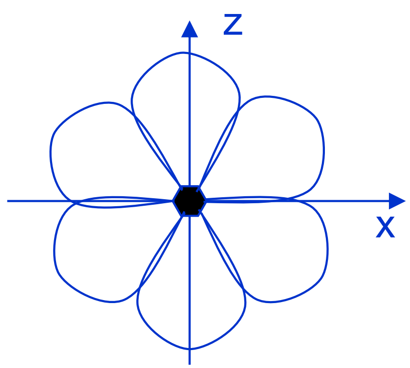


top view (xz-plane)

directed
antenna



top view, 3 sector



top view, 6 sector

sectorized
antenna

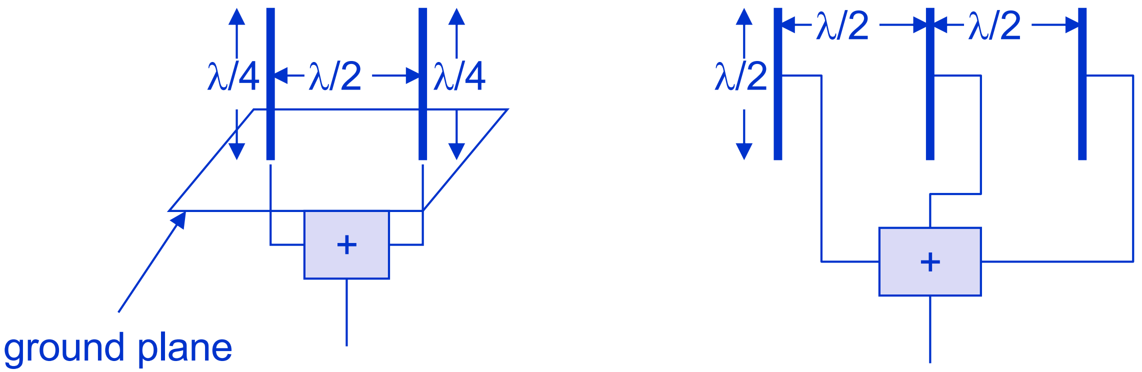
Antennas: diversity

Grouping of 2 or more antennas

- multi-element antenna arrays

Antenna diversity

- switched diversity, selection diversity
  - receiver chooses antenna with largest output
- diversity combining
  - combine output power to produce gain
  - cophasing needed to avoid cancellation



Channel Capacity – Signal vs. Interference

Questions to be answered:

1. How do interference and the received signal strength determine the capacity of a channel?
2. Why is interference – or more acurrate its absence – as important as the signal itself?
3. What defines an interference-limited system? What happens there? How can we increase the system capacity in these systems? And how not?

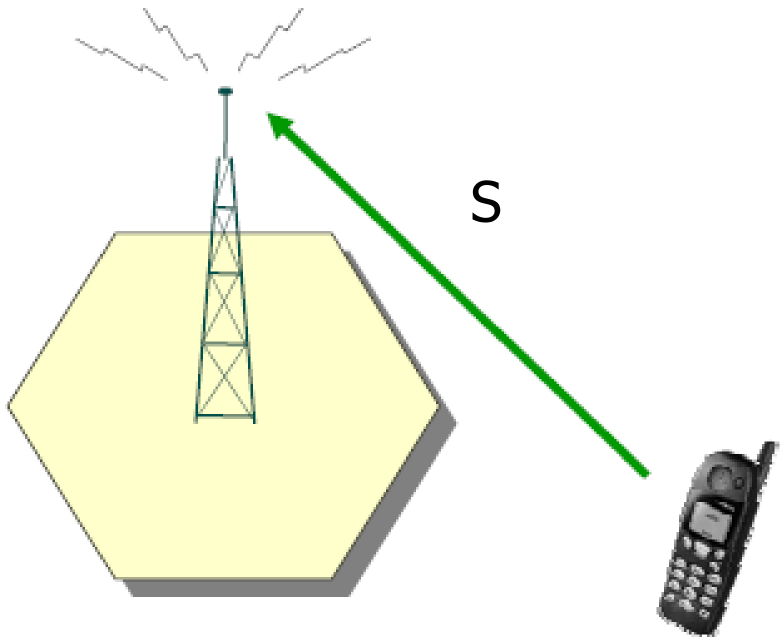
Range-limited systems (lack of coverage)

- Mobile stations located far away from BS (at cell border or even beyond the coverage zone)

- S at the receiver is too low (below receiver sensitivity) because the path loss between sender and receiver is too high

➔ S is too low

➔ No signal reception possible



Quelle: B. Walke, M.P. Althoff, P.Seidenberg, UMTS – Ein Kurs, Weil der Stadt 2001

Interference-limited systems (lack of capacity)

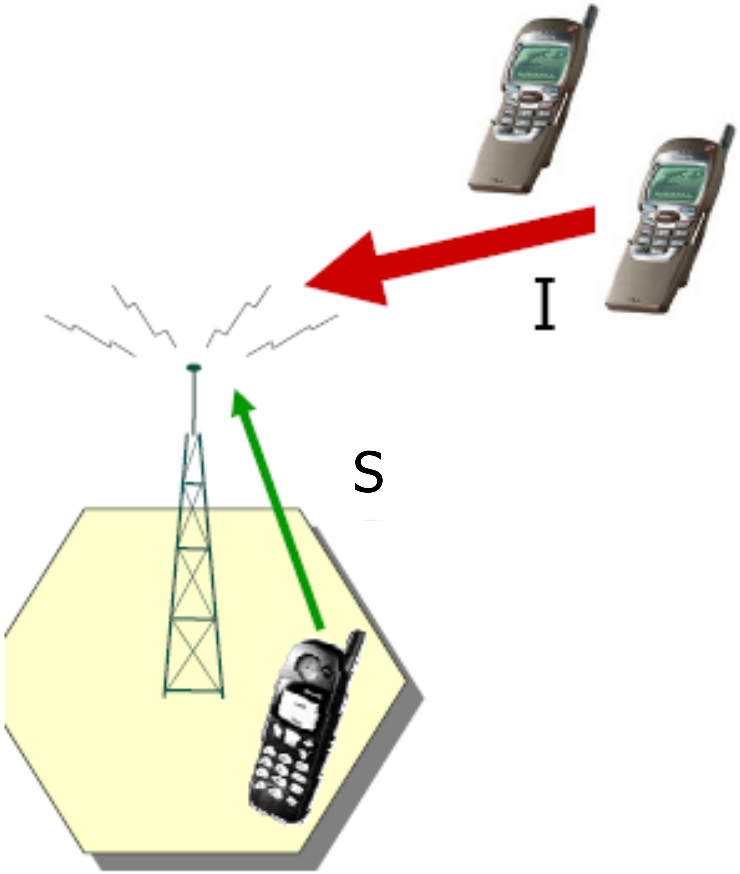
Picture shows uplink communication!

- Mobile station is within coverage zone
- S is sufficient, but too much interference I at the receiver

➔ SINR is too low

➔ No more resources / capacity left

For a high system capacity, the signal strength is as important as the abscence of high levels of interference!



Quelle: B. Walke, M.P. Althoff, P.Seidenberg, UMTS – Ein Kurs, Weil der Stadt 2001

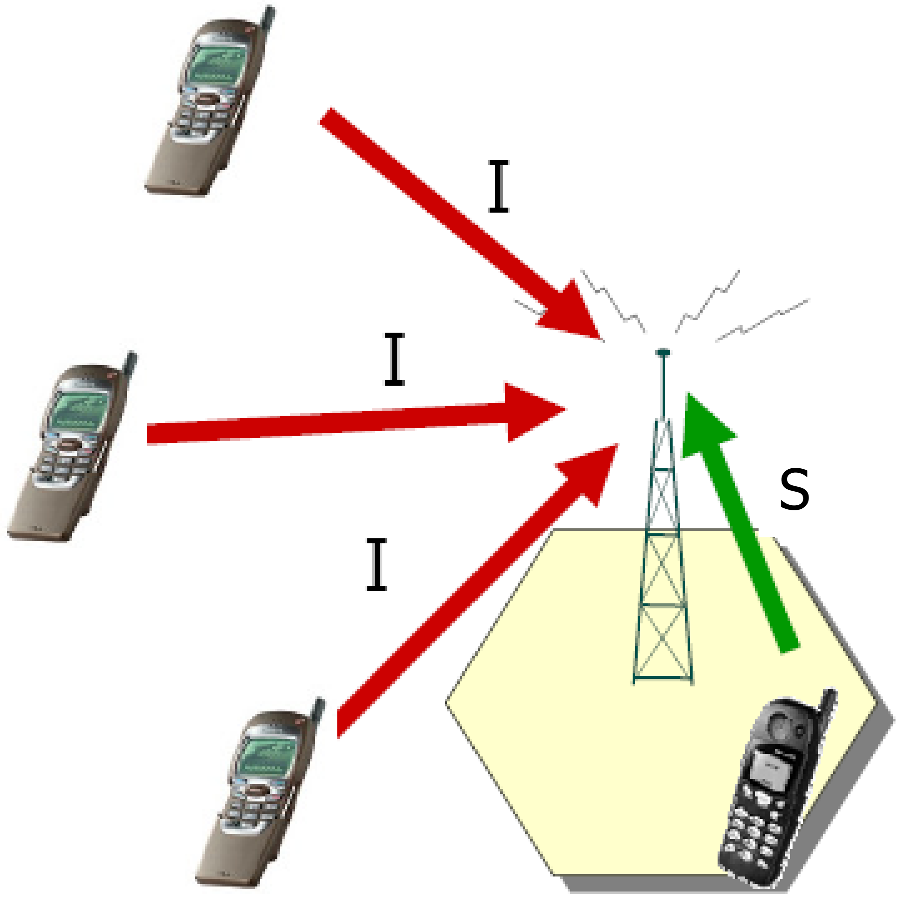
Signal to Interference and Noise Ratio (SINR)

Ratio of Signal-to-Interference-and-Noise power at the receiver

$$SINR = \frac{S}{\sum I_j + N}$$

- the SINR influences the error rate of the received signal
- the minimum required SINR depends on the system and the signal processing capabilities of the receiver technology
- in cellular systems interference typically dominates noise
- typical required SINR value for GSM is 15dB (Factor 32)
- highly coded signals, e.g. UMTS, work with much lower SINR requirements

Picture shows uplink communication only!

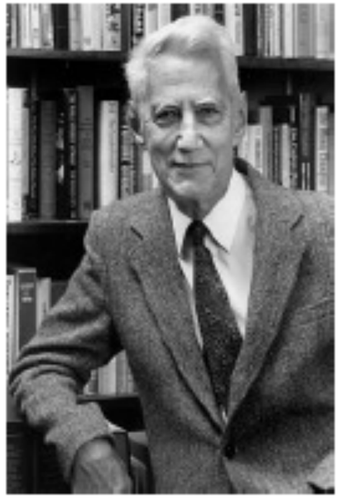


Quelle: B. Walke, M.P. Althoff, P.Seidenberg, UMTS – Ein Kurs, Weil der Stadt 2001,

Information Theory: Channel Capacity (1)

- Bandwidth limited Additive White Gaussian Noise (AWGN) channel
- Gaussian codebooks
- Single transmit antenna
- Single receive antenna (SISO)

- Shannon (1950):
  Channel Capacity <=
  Maximum mutual information between sink and source



C. Shannon
Bell Labs Technical Journal, 1950

Tx

$$C = B \log_2\left(1 + \frac{S}{N}\right) \text{ [bps]}$$

Rx

Channel Capacity

Bandwidth

Signal-to-noise ratio SNR

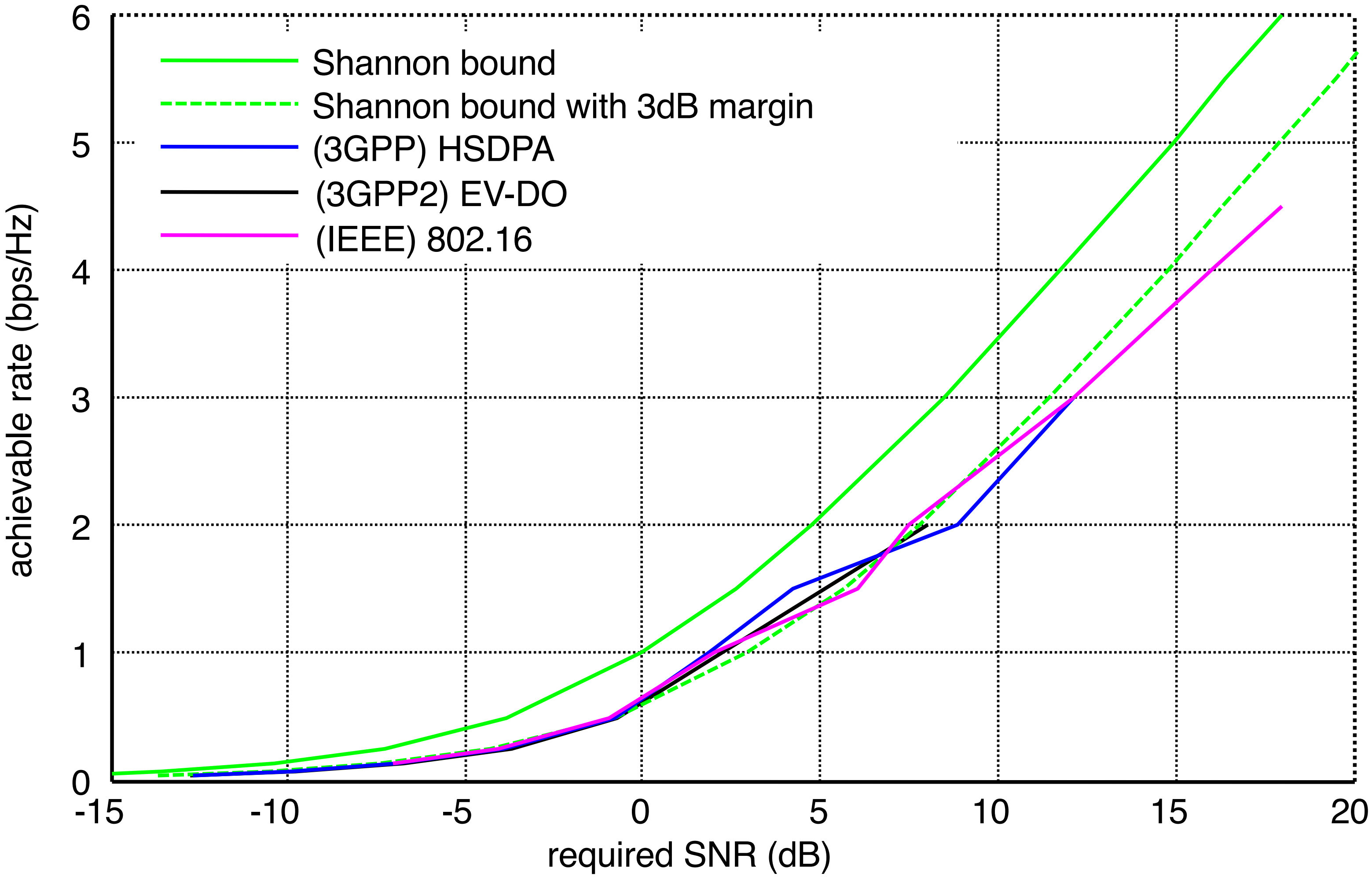
Information Theory: Channel Capacity (2)

- For S/N >>1 (high signal-to-noise ratio), approximate

$$C \approx B \, \frac{1}{3} \, \frac{S}{N}_{dB} \qquad \text{[bps]}$$

- **Observation: Bandwidth and S/N are reciprocal to each other**
- This means:
  - with low bandwidth very high data rate is possible, provided
    - S/N is high enough
    - e.g. allowing the use of higher order modulation schemes
  - with high noise (low S/N) data communication is possible if
    - bandwidth is large
    - e.g. employing spread spectrum technology

Link Capacity for Various Rate-Controlled Technologies



- Link capacity of systems is approaching the Shannon limit (within a factor of two)
- Improvements in spectral efficiency focus on intelligent antenna techniques and/or improved coordination between base stations

Link performance of OFDM & 3G systems are similar and approaching the (physical) Shannon bound

Multiplexing – Splitting up the Radio Resource

Questions to be answered:

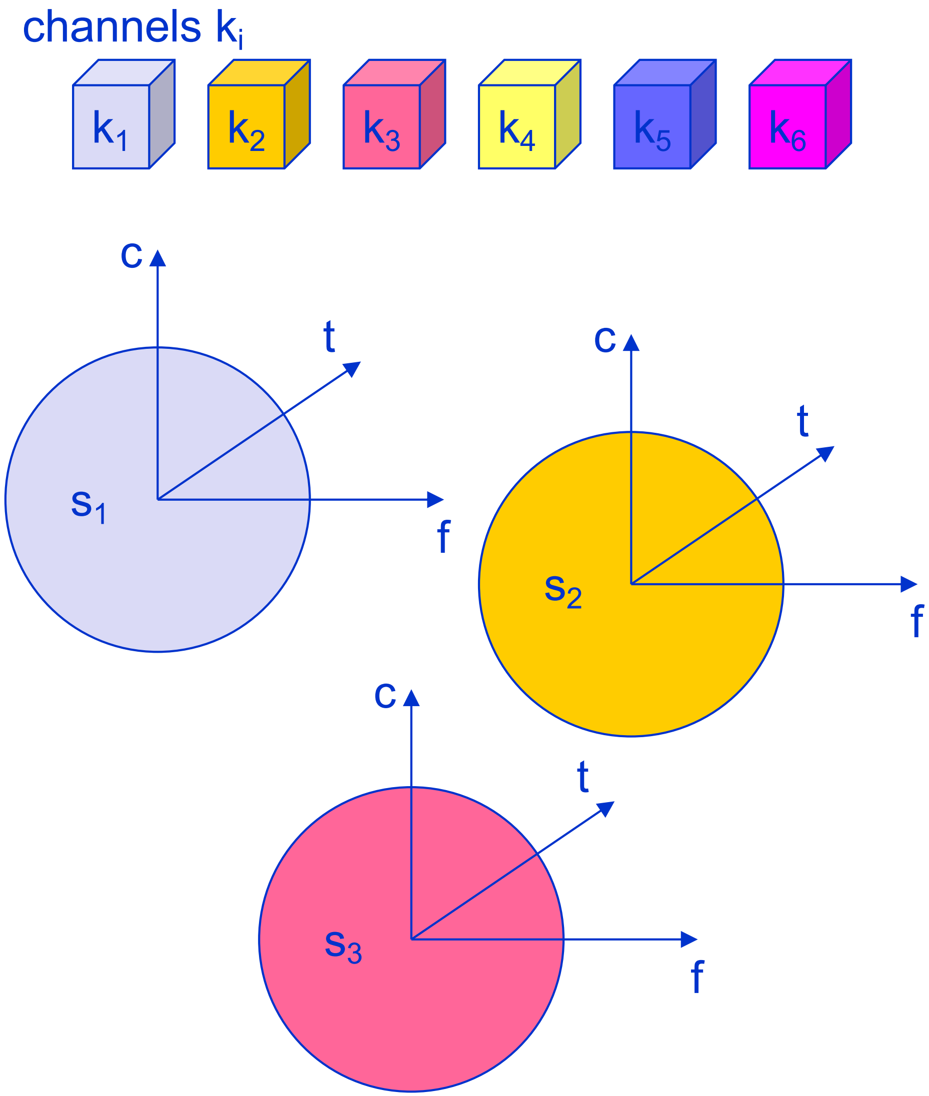
1. What is the purpose of multiplexing?
2. What kind of multiplexing is used for public FM radio?
3. Which kind of multiplexing is used for GSM?
4. What is the problem of multiplexing in time, what is its advantage?

Multiplexing

Goal: multiple use of a shared medium to provide it to multiple entities rather than just a single one

Multiplexing in 4 dimensions

- space (s_i)
- time (t)
- frequency (f)
- code (c)



Important: guard spaces needed!

Frequency multiplex

Separation of the whole spectrum into smaller frequency bands

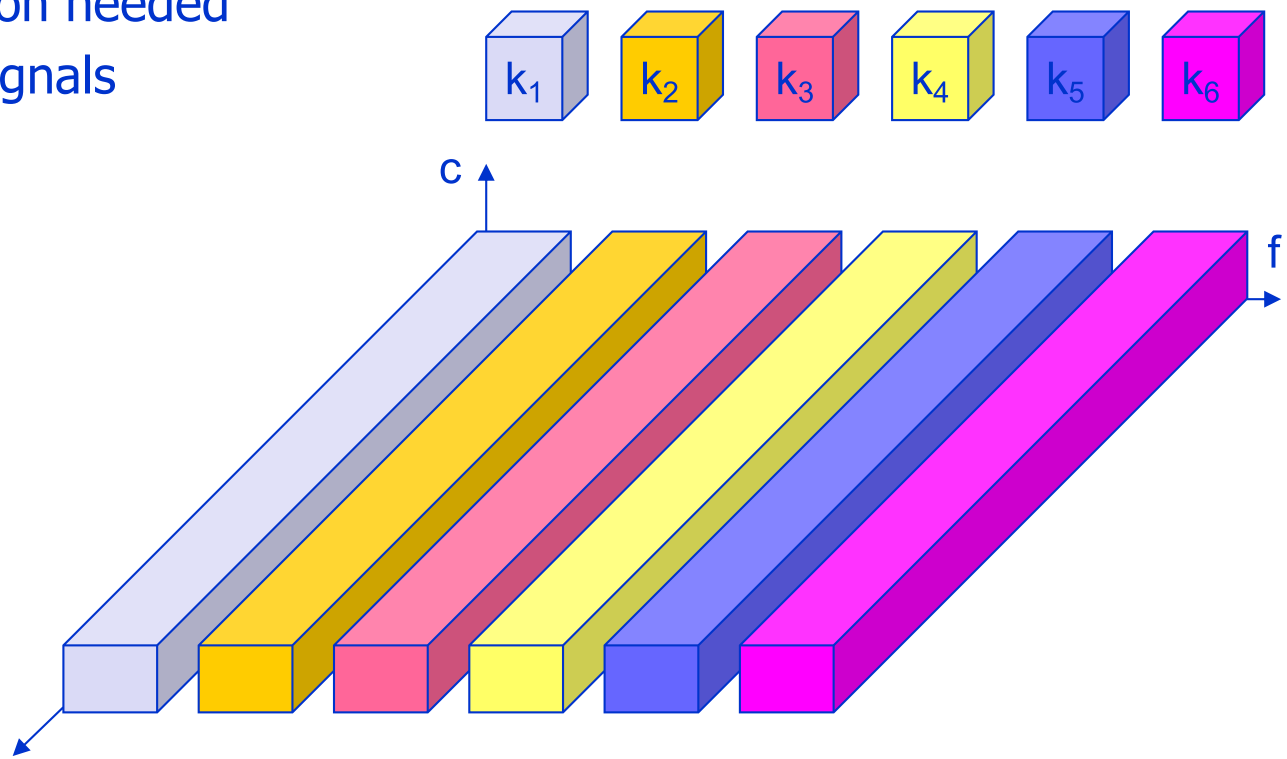
A channel gets a certain band of the spectrum for the whole time

Advantages:

- no dynamic coordination needed
- applicable to analog signals

Disadvantages:

- waste of bandwidth if the traffic is distributed unevenly
- inflexible
- guard space



Time multiplex

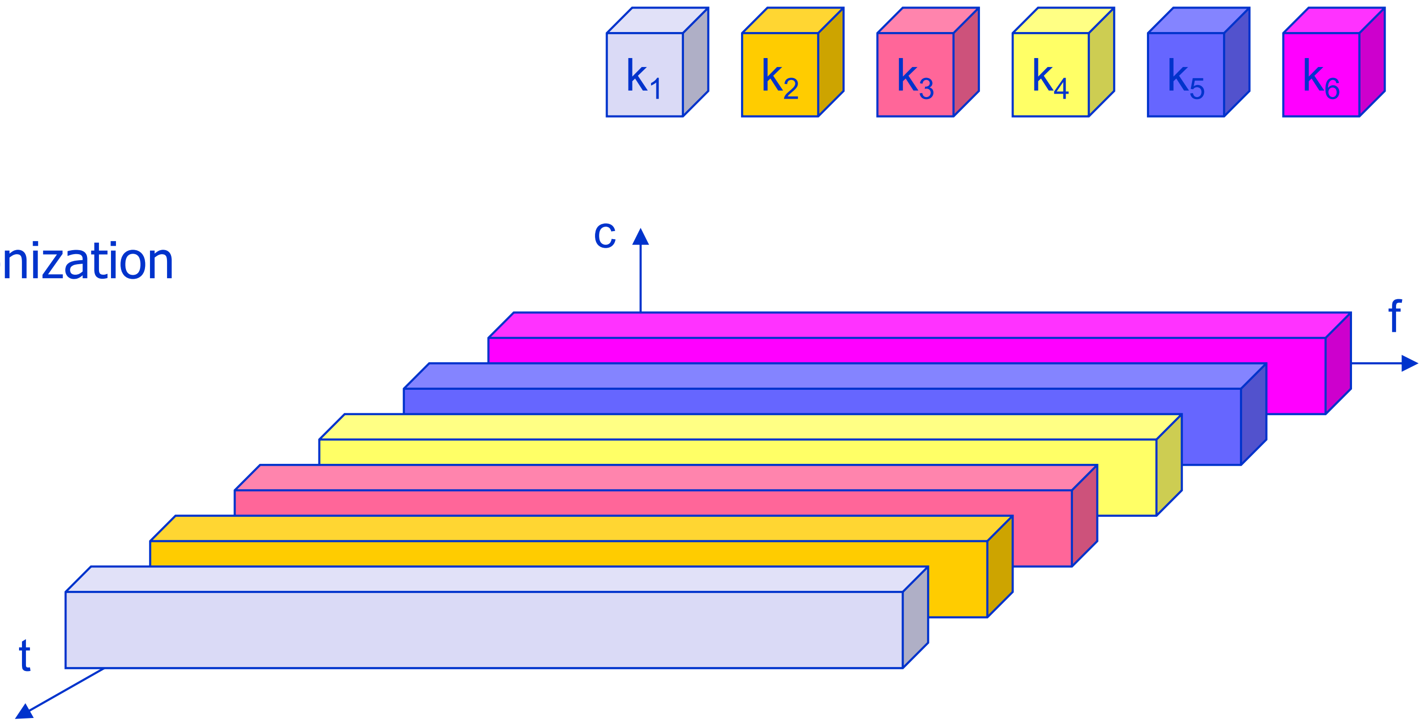
A channel gets the whole spectrum for a certain amount of time

Advantages:

- only one carrier in the medium at any time
- throughput high even for many users

Disadvantages:

- precise synchronization needed



Time and frequency multiplex

Combination of both methods

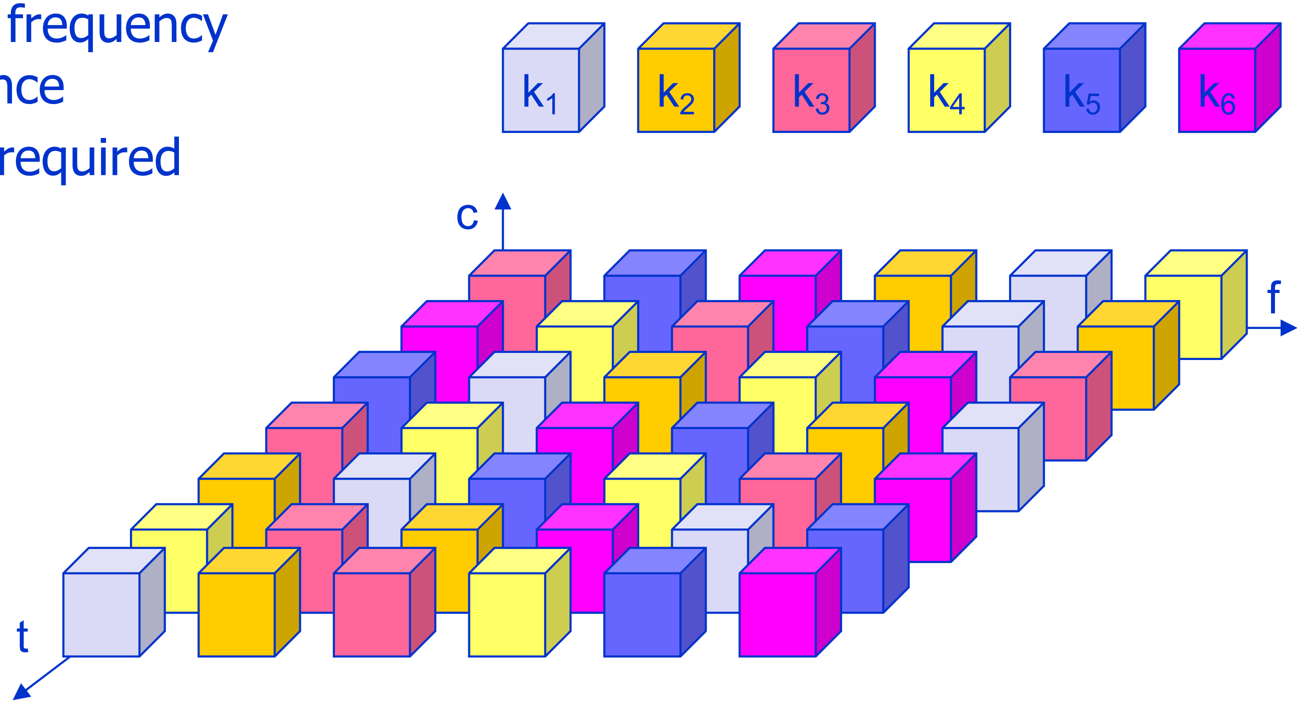
A channel gets a certain frequency band for a certain amount of time

Example: GSM (frequency hopping)

Advantages:

- some (weak) protection against tapping
- protection against frequency selective interference

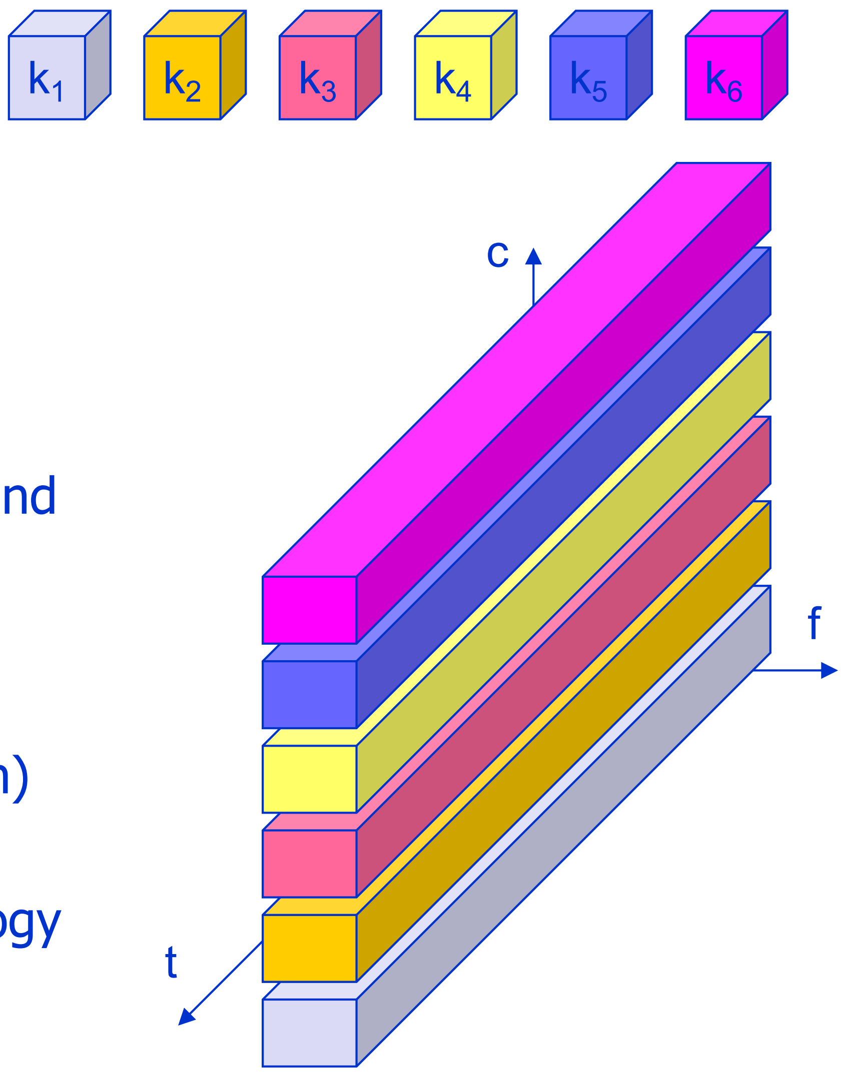
but: precise coordination required



Code multiplex

Each channel has a unique code

All channels use the same spectrum at the same time



Advantages:

- bandwidth efficient
- no coordination and synchronization necessary
- good protection against interference and tapping

Disadvantages:

- complex receivers (signal regeneration)

Implemented using spread spectrum technology

Modulation and Coding

Questions to be answered:

1. How can we add information to a periodic signal? What are the choices?
2. Why is amplitude modulation in its pure form rarely used in practice? When does it make sense? Where is the idea applied and what is the benefit of its application?
3. What is the advantage of 16QAM over QPSK?
4. What limits the use of higher order modulation schemes, as e.g. 16QAM or 64QAM?
5. How can we optimize the throughput of a single transmitter-receiver pair? How can we optimize it under given bandwidth constraints as well as fixed signal and interference values?

Modulation

Basic schemes

- Amplitude Modulation (AM)
- Frequency Modulation (FM)
- Phase Modulation (PM)

Motivation for modulation

- smaller antennas (e.g., $\lambda/4$)
- Frequency Division Multiplexing
- medium characteristics
- spectrum availability

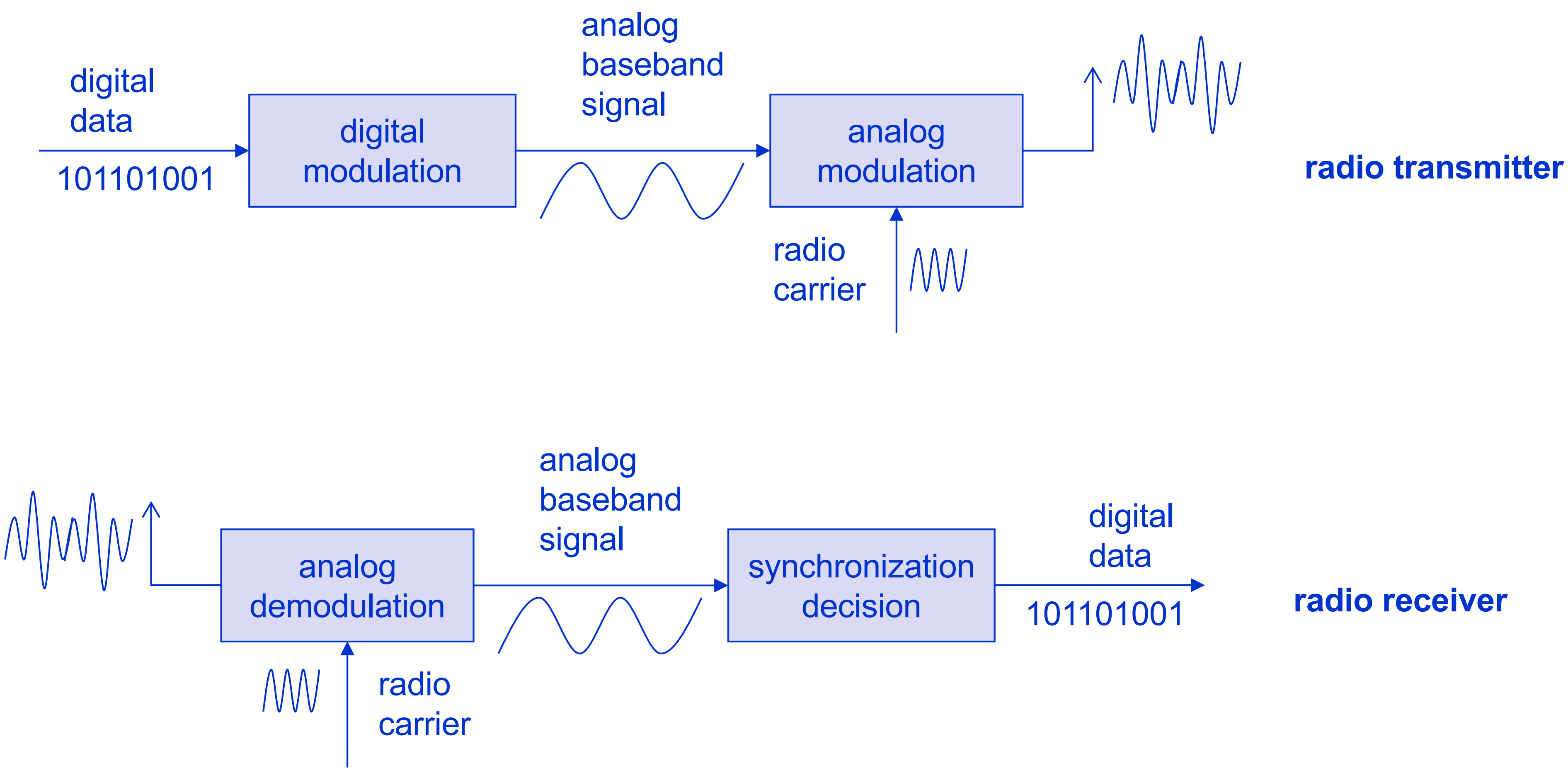
Analog modulation

- shifts center frequency of baseband signal up to the radio carrier

Digital modulation

- digital data is translated into an analog signal (baseband)
- ASK, FSK, PSK (see next slides)
- differences in spectral efficiency, power efficiency, robustness

Modulation and demodulation



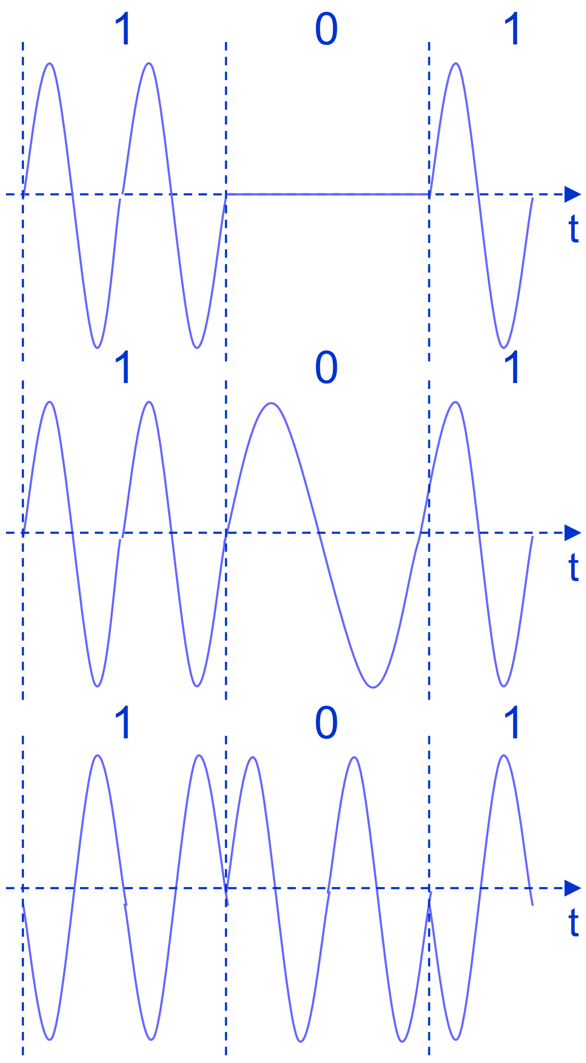
Digital modulation

Modulation of digital signals known as Shift Keying

Amplitude Shift Keying (ASK):
- very simple
- low bandwidth requirements
- very susceptible to interference

Frequency Shift Keying (FSK):
- needs larger bandwidth

Phase Shift Keying (PSK):
- more complex
- robust against interference

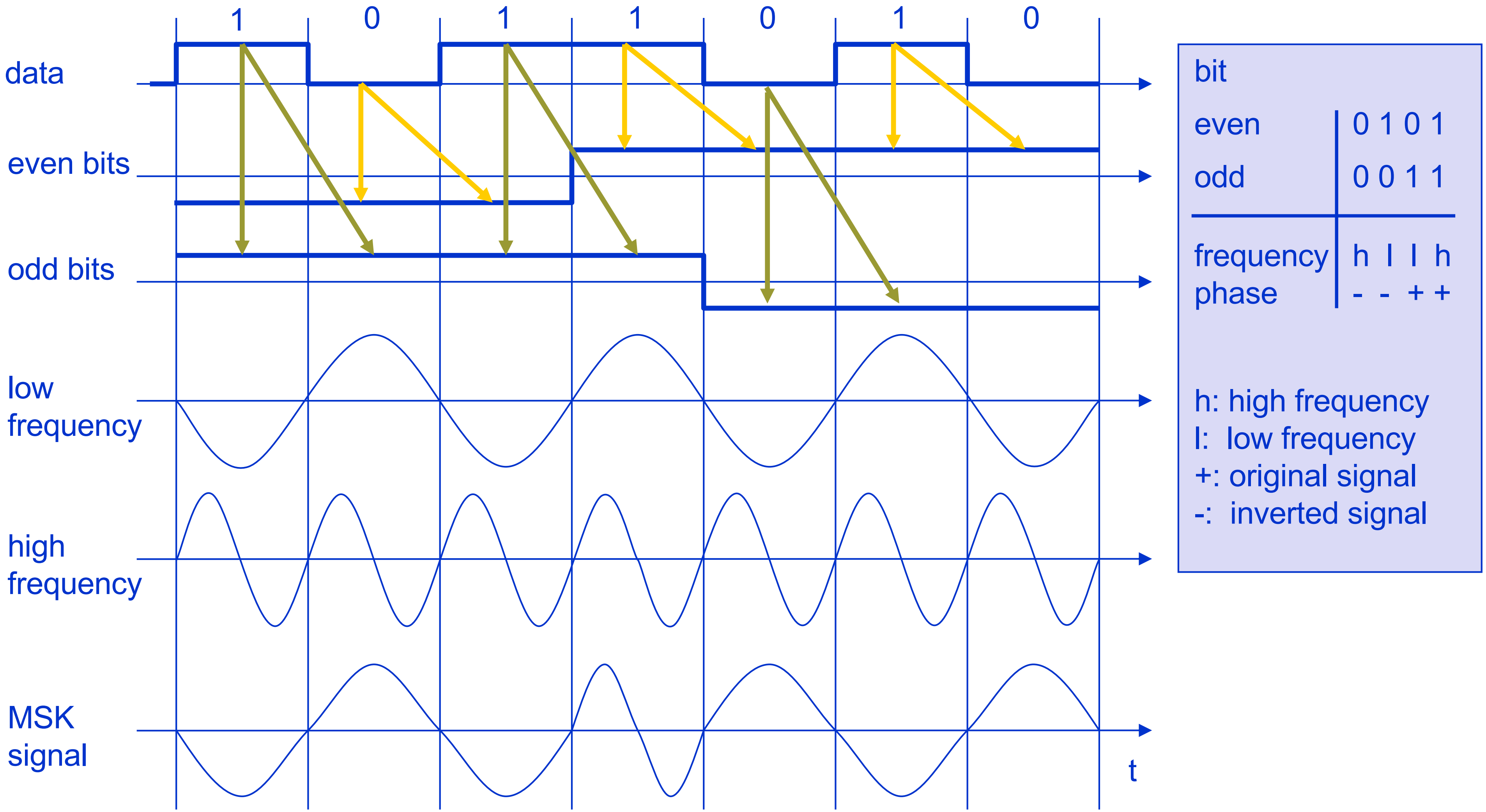


Advanced Frequency Shift Keying

- bandwidth needed for FSK depends on the distance between the carrier frequencies
- special pre-computation avoids sudden phase shifts
  ➔ Continuous Phase Modulation (CPM)
  e.g. MSK (Minimum Shift Keying)
- bit stream is separated into even and odd bits, the duration of each bit is doubled
- depending on the bit values (even, odd) the higher or lower frequency, original or inverted is chosen
- the frequency of one carrier is twice the frequency of the other, eliminating abrupt phase changes

- even higher bandwidth efficiency using a Gaussian low-pass filter
  ➔ GMSK (Gaussian MSK), used for GSM and DECT

Example of MSK

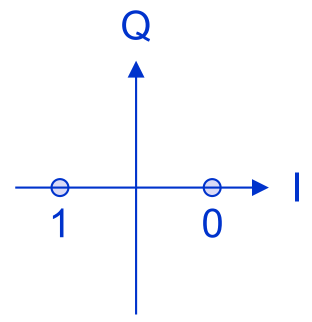


MSK avoids sudden phase shifts!

Advanced Phase Shift Keying

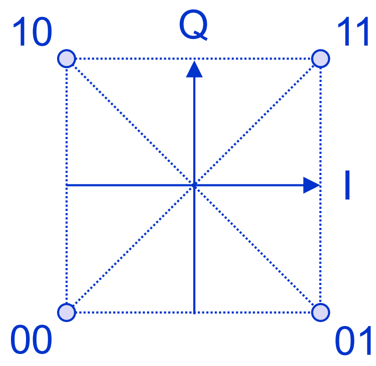
BPSK (Binary Phase Shift Keying):

- bit value 0: sine wave
- bit value 1: inverted sine wave
- very simple PSK
- low spectral efficiency
- robust, used e.g. in satellite systems



QPSK (Quadrature Phase Shift Keying):

- 2 bits coded as one symbol
- symbol determines shift of sine wave
- needs less bandwidth compared to BPSK
- more complex
- used in UMTS
- often also transmission of relative, not absolute phase shift: DQPSK - Differential QPSK (IS-136, PHS)

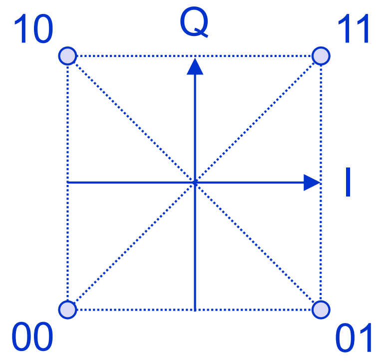


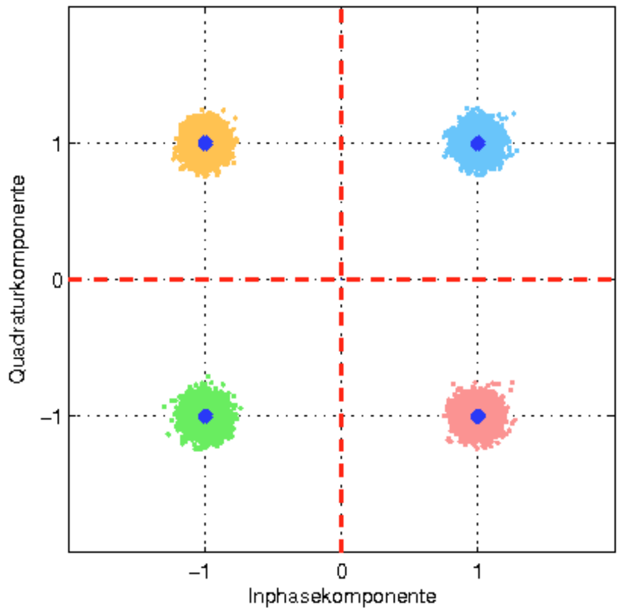
Pulse filtering of baseband to avoid sudden phase shifts

=> reduce bandwidth of modulated signal

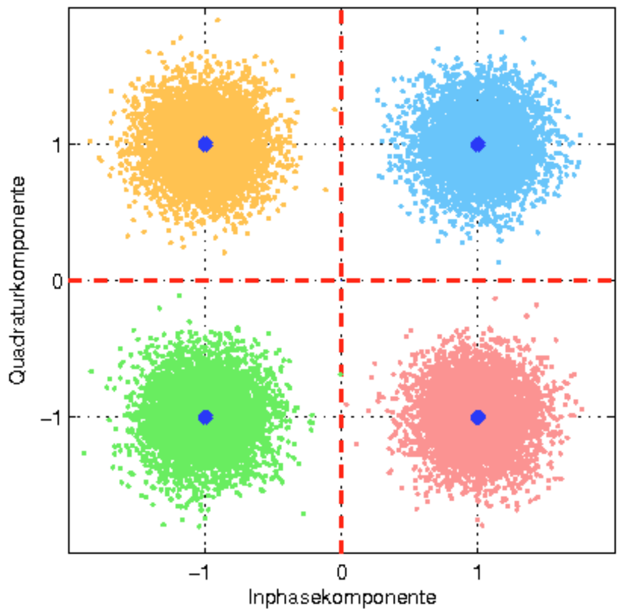
Advanced Phase Shift Keying

QPSK with different noise levels (low to high)

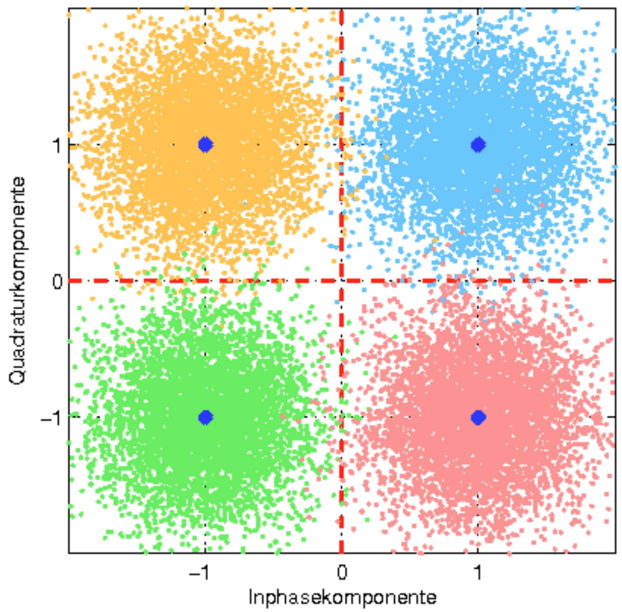




inefficient resource use, waste of resource



perfect, small number of errors/retransmissions



inefficient, too many errors/ retransmissions

Quadrature Amplitude Modulation

Quadrature Amplitude Modulation (QAM)

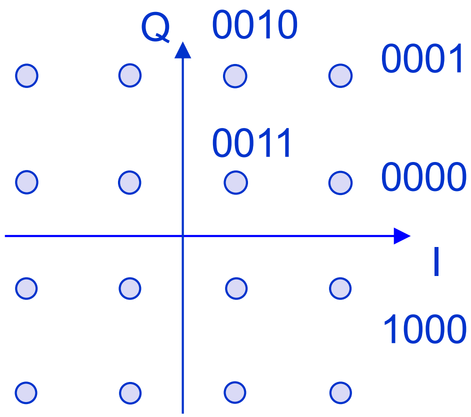
- combines amplitude and phase modulation
- it is possible to code n bits using one symbol
- 2^n discrete levels, n=2 identical to QPSK
- bit error rate increases with n, but less errors compared to comparable PSK schemes

Example: 16-QAM (4 bits = 1 symbol)

Symbols 0011 and 0001 have the same phase, but different amplitude

0000 and 1000 have different phase, but same amplitude

QAM is widely used, e.g. in LTE, GPRS, WLAN



Channel coding

Idea: Data are superimposed with some code before transmission, adding redundancy to the data stream; this permits the receiver to detect or even tolerate and recover from some sporadic transmission errors which is called Forward Error Correction (FEC).

Of course, highly redundant code adds overhead and thus, increases the data volume to be transmitted.

Thus, there is a tradeoff between

- the chance of recovery from potential errors without retransmisson, and
- the overhead needed for this for the case that there was no error.

An alternative to redundant coding are explicit retransmissions (ARQ) in the case of errors.

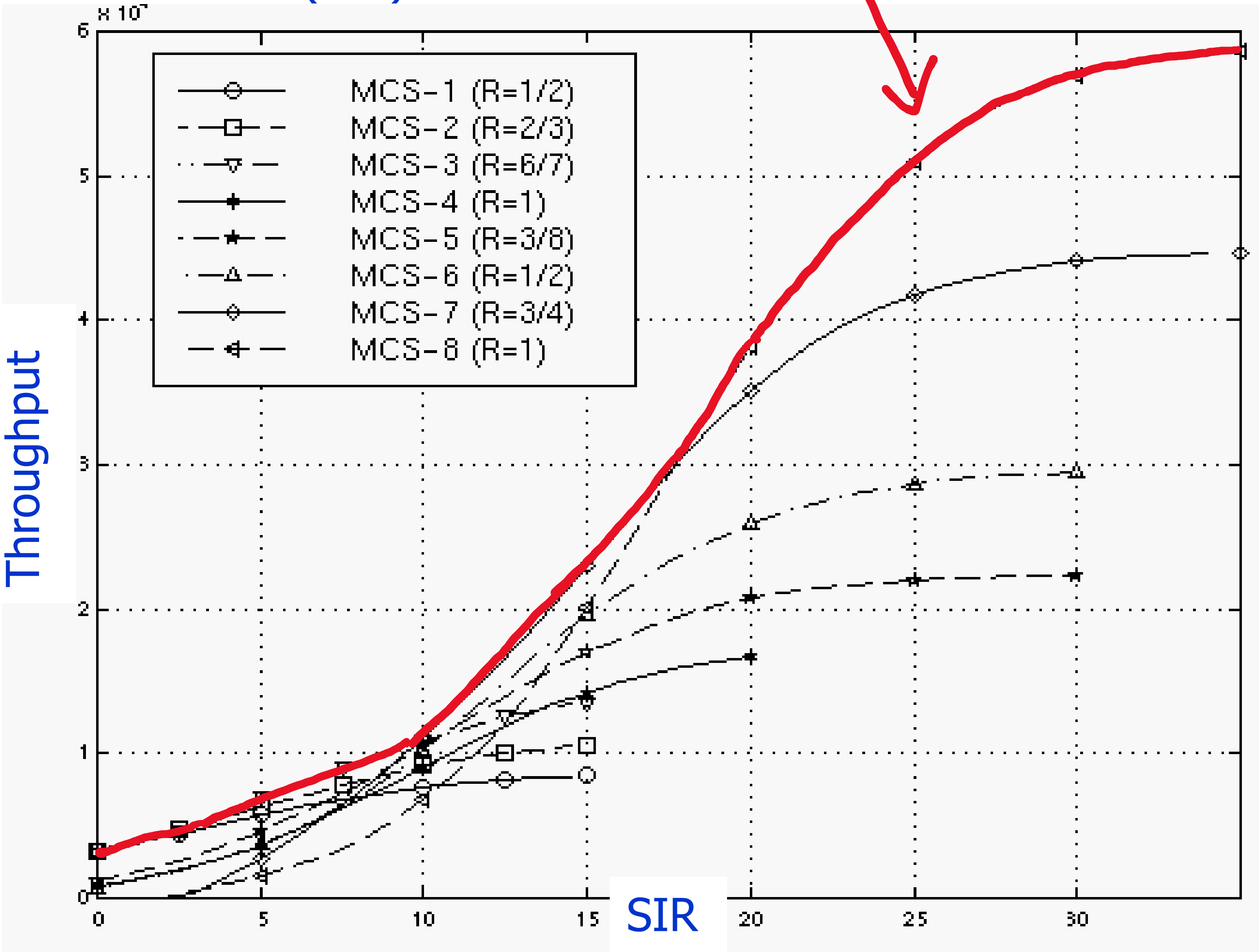
Modulation: Channel coding is typically applied in combination with a modulation scheme to best adapt to the channel characteristics and thus, to maximize the data throughput.

Interference: Channel coding can be used as well to simplify the separation between the user signal and interference from others, e.g. as done in UMTS/CDMA systems

Note that channel coding differs from source coding, which is focusing on the efficient coding of information from different sources, e.g. text, voice or video.

Example for Adaptive Modulation and Coding in EDGE

Scheduler selects the coding scheme that provides maximum throughput for given channel condition (SIR)



Spectrum Spreading

Questions to be answered:

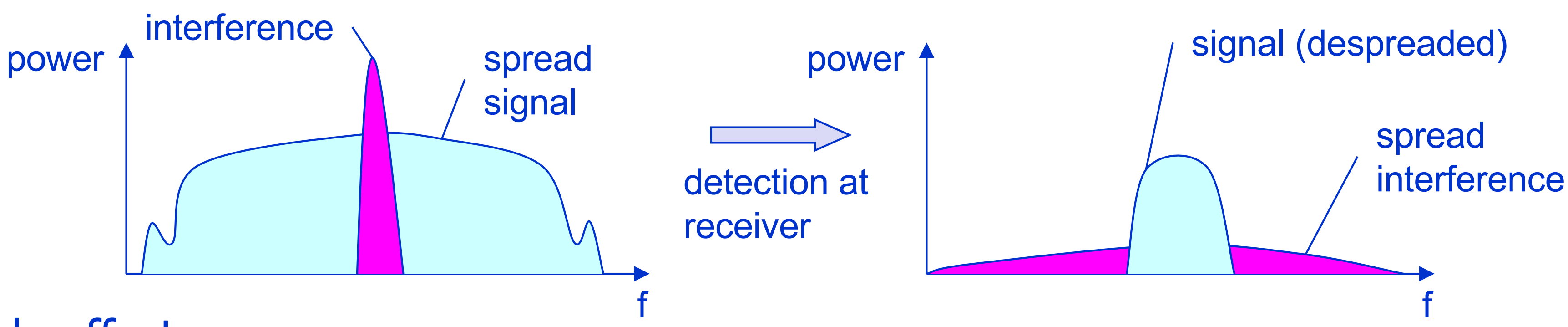
1. What is the benefit of spectrum spreading? What is needed for spreading?
2. What is the benefit of frequency hopping? What are its problems?

Spread spectrum technology

Problem of radio transmission: frequency dependent fading can wipe out narrow band signals for duration of the interference

Solution: spread the narrow band signal into a broadband signal using a special code

⇒ protection against narrow band interference



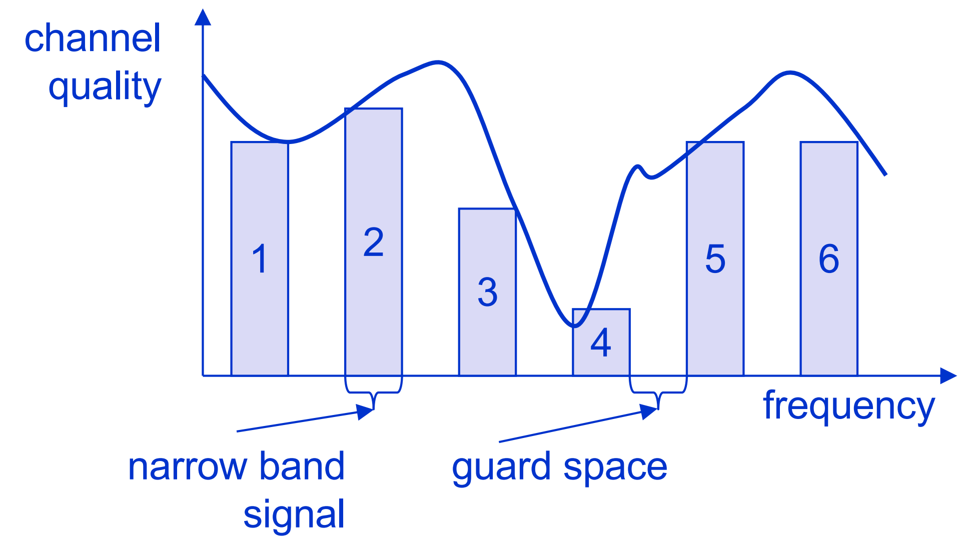
Side effects:

- coexistence of several signals without dynamic coordination
- tap-proof

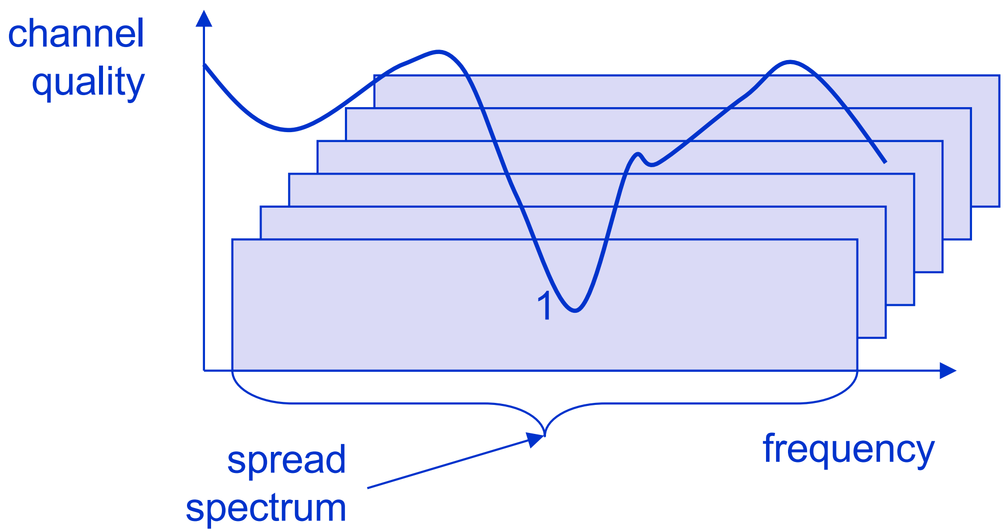
Alternatives:

- Direct Sequence (UMTS)
- Frequency Hopping (slow FH: GSM, fast FH: Bluetooth)

Spreading and frequency selective fading



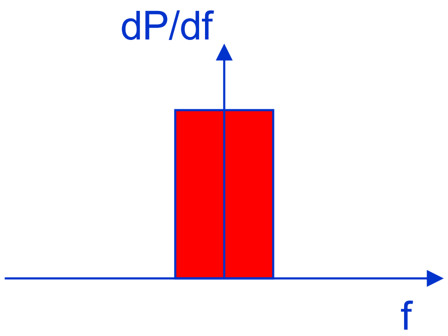
narrowband interference without spread spectrum



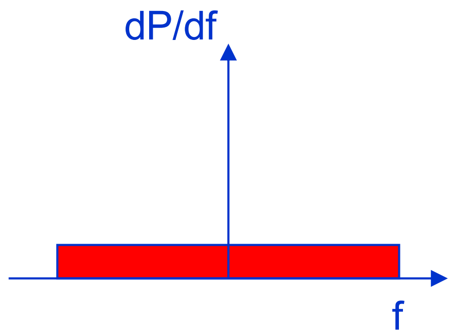
spread spectrum to limit narrowband interference

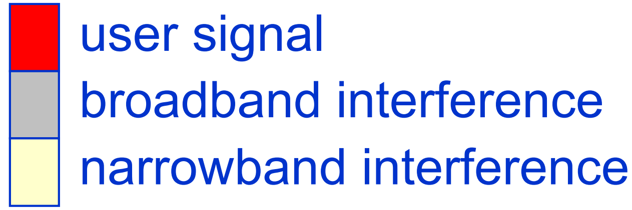
Effects of spreading and interference

i) narrow band signal



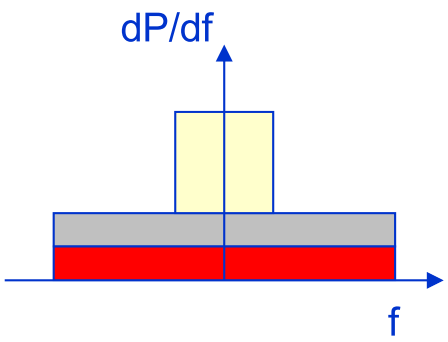
ii) spreaded signal (broadband signal)



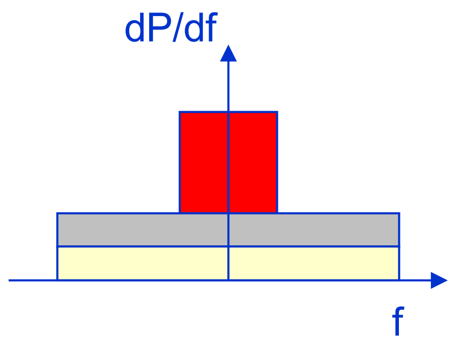


sender

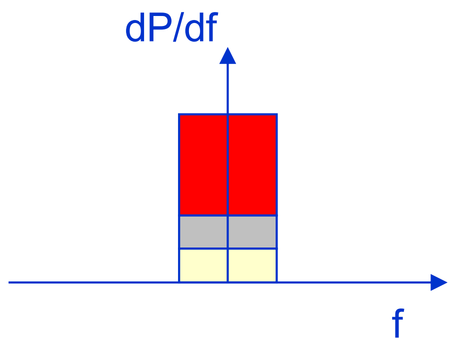
iii) addition of
interference



iv) despreaded
signal



v) application of
bandpass filter



receiver

DSSS (Direct Sequence Spread Spectrum)

XOR of the signal with pseudo-random number (chipping sequence)

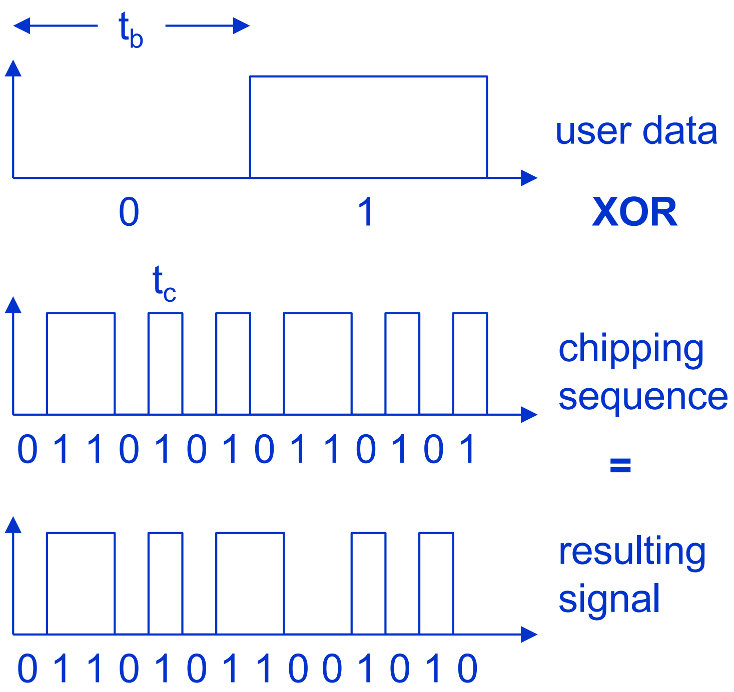
- many chips per bit (e.g., 128) result in higher bandwidth of the signal

Advantages

- reduces frequency-selective fading
- in cellular networks
  - all base stations can use the same frequency range
  - several base stations can detect and recover the signal
  - soft handover

Disadvantages

- precise power control needed



t_b: bit period
t_c: chip period

FHSS (Frequency Hopping Spread Spectrum) I

Discrete changes of carrier frequency

- sequence of frequency changes determined via pseudo random number sequence

Two versions

- Fast Hopping:
  several frequencies per user bit
- Slow Hopping:
  several user bits per frequency

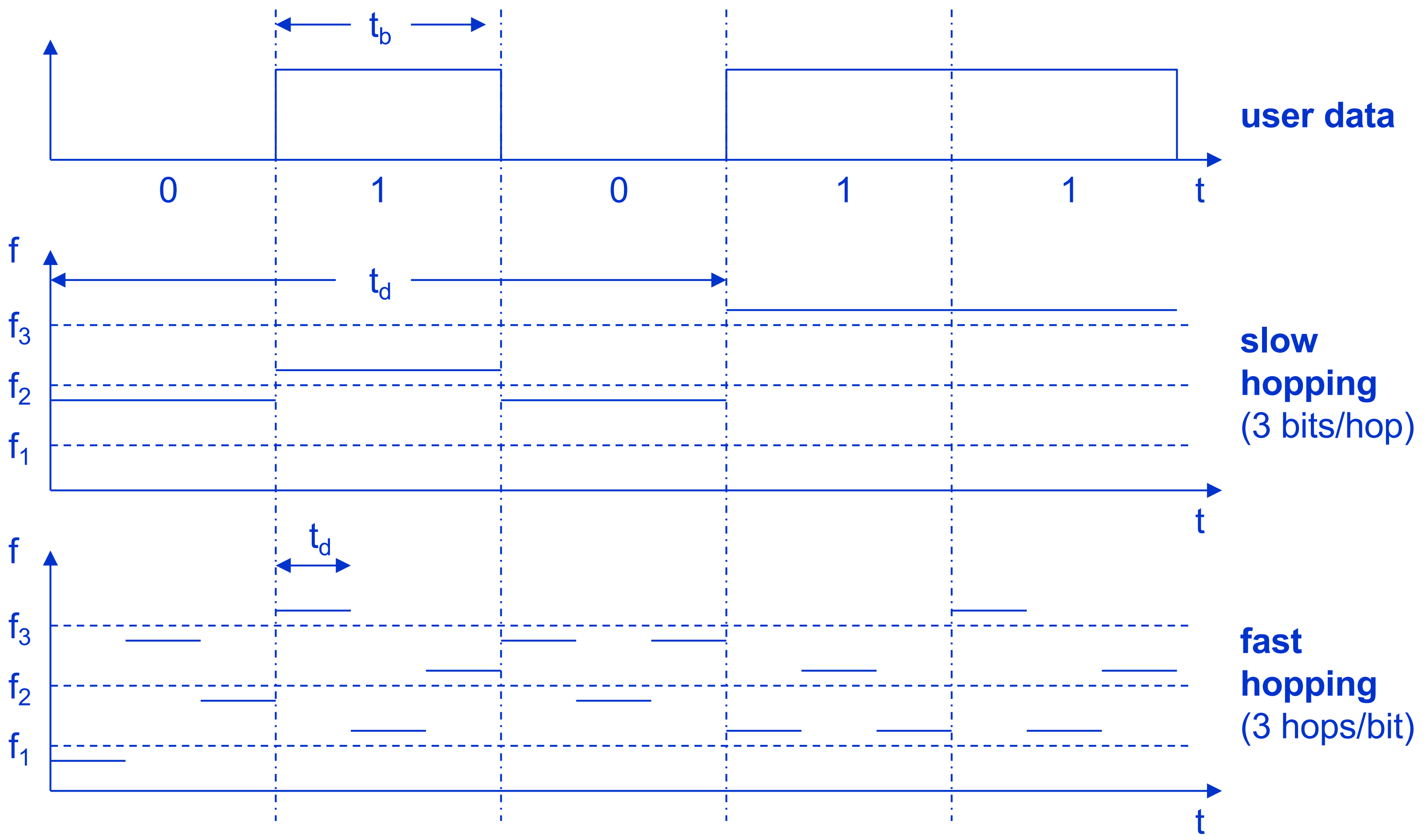
Advantages

- frequency selective fading and interference limited to short period
- simple implementation
- uses only small portion of spectrum at any time

Disadvantages

- not as robust as DSSS
- simpler to detect

FHSS (Frequency Hopping Spread Spectrum) II



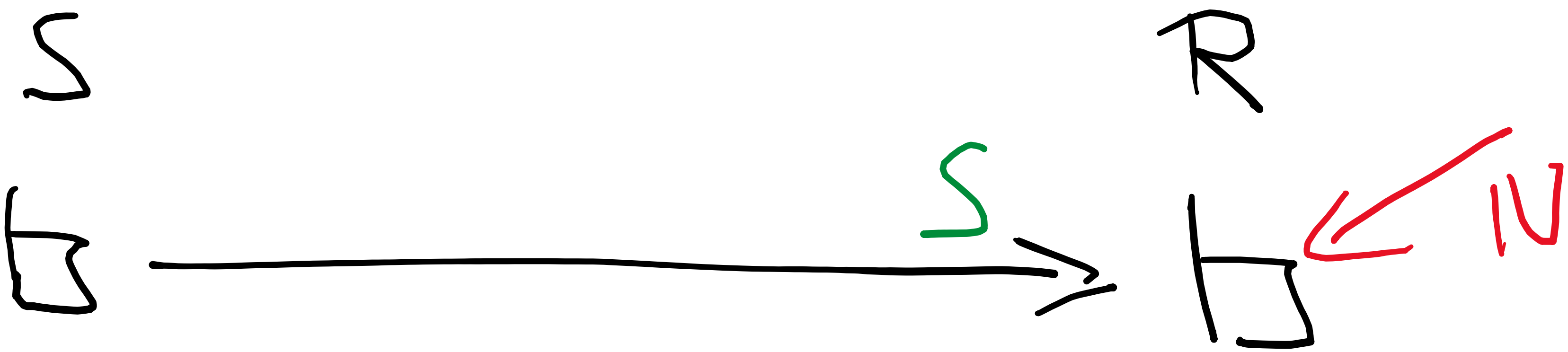
t_b: bit period t_d: dwell time

Cellular Networks – Capacity and Interference

Questions to be answered:

1. Spatial reuse in cellular systems: compare a reuse of 3 with a reuse of 9 with respect to capacity (data rate) and quality (error rate).
2. What is the effect of increasing the transmit power of all base stations in interference-limited systems?
3. How can we increase the system capacity in interference-limited cellular systems? And how not? Discuss the effects of different approaches.
4. What are the results of adding more base stations to a given cellular network layout assuming that there is no extra spectrum available? What are the measures that need to be taken to handle this?
5. What are the effects of increasing the spatial separation, e.g. employing 6 instead of 3 sectors per base station?
6. Discuss how direct communication between mobiles (D2D) is a solution to improve the network capacity of cellular networks? What are the gains of this?

Review: single sender-receiver pair



Given are:
- Channel bandwidth B
- Noise level N

How can we maximize the transmitted data volume?

Remember S/N in Shannons formula?
$\Rightarrow$ Maximize the transmit power!

Ok, but there are limits! What happens if the transmit power is reached?
What else can we do to get the maximum out of our channel?

Remember: we can optimize
- modulation (BPSK or 16-QAM?)
- coding (zero redundancy or factor of 5 redundancy?)

$\Rightarrow$ Find best combination of modulation and coding scheme

Cellular systems: cell structure

Implements space division multiplex:

- base station covers a certain transmission area (cell)

Mobile stations communicate via the base station only

Advantages of cell structures:

- higher capacity, higher number of users
- less transmission power needed
- more robust, decentralized
- base station deals with interference, transmission area, etc. locally

Problems:

- Infrastructure side:
  - lots of base stations needed
  - fixed network needed connecting the base stations
- Layer 2 and 3: handover (changing from one cell to another) necessary
- Layer 1: interference with other cells

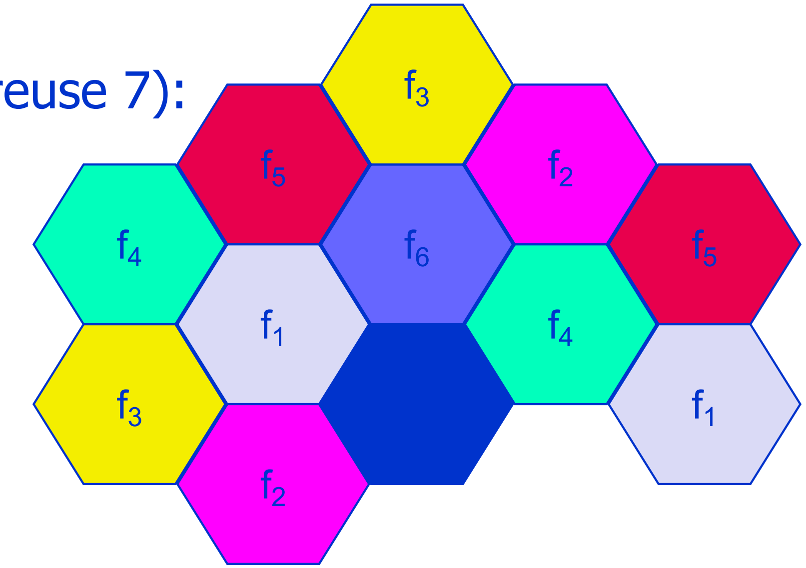
Cell sizes vary from 10s of meters in urban areas to many km in rural areas (e.g. maximum of 35 km radius in GSM)

Cellular systems: frequency planning I

Frequency reuse between base stations only beyond a certain distance

$\Rightarrow$ Minimize interference among cells

Standard (hexagon) model using 7 frequencies (reuse 7):



Fixed frequency assignment:

- certain frequencies are assigned to a certain cell
- problem: different traffic load in different cells

Dynamic frequency assignment:

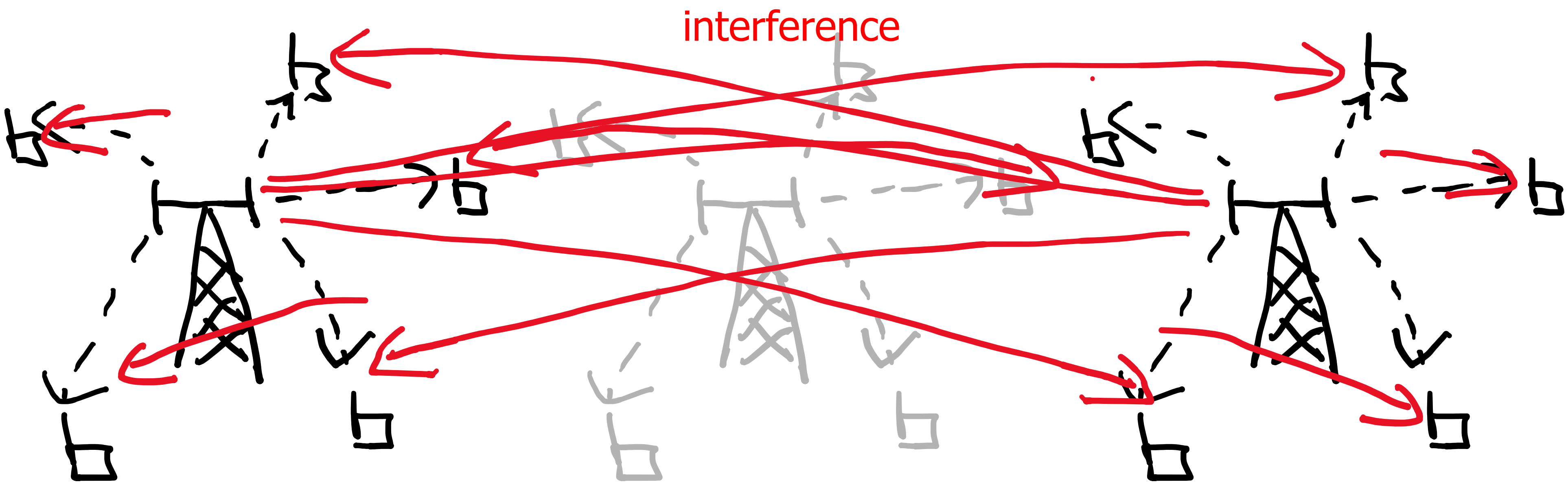
- base station chooses frequencies depending on the frequencies already used in neighbor cells
- more capacity in cells with more traffic
- assignment can also be based on interference measurements

The interference problem in cellular systems

First cell

Second cell

Third cell



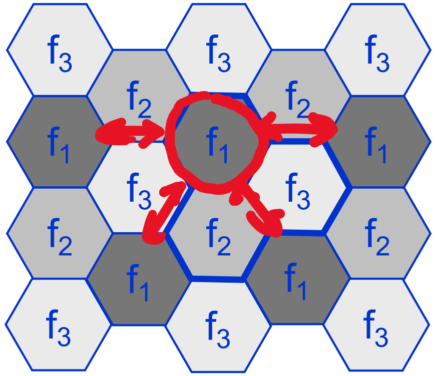
Frequency 1

Frequency 2 (different)
Not relevant for
Interference situation!

Reuse of frequency 1!

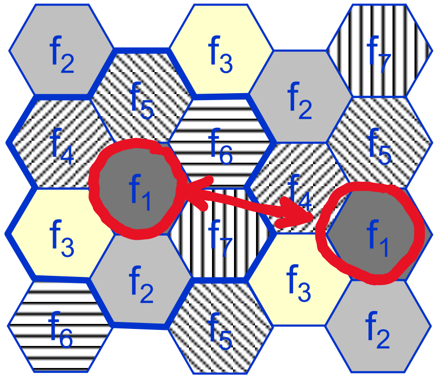
Generates interference
on first cell!
And vice versa!

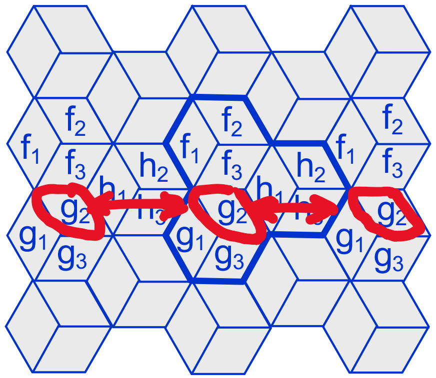
Cellular systems: frequency planning II



3 cell cluster
(reuse 3)

7 cell cluster
(reuse 7)





3 cell cluster
with 3 sector antennas
(reuse 9)

Wrap-up

Understanding of the signal-interference situation at the receiver is the key to the understanding of mobile communication networks and their planning!

The global goal of cellular networks, e.g. of LTE systems, is the maximization of the capacity of the system as a whole; this requires understanding of

- the signal (from sender), with pathloss and multi-path issues => fading
- interference from others using the same radio resource (typically from other cells)

Altogether the S/I relation defines the capacity of each receicer/comm. link, thus, we have a high dependence on the interferers, which turns the problem into a complex optimization problem.

Lessons to take home:

- We have IF-limited systems => Considering interference is as important as the signal
- There are pros and cons of the increase of the transmit power/signal strength!
- Global goal is to optimize the S/I situation for each and all communications
- As pathloss is at least d^2 (often d^4), the minimization of the distance to the receiver results in dramatic gains in system capacity
- There is no free lunch: every design decision has its pros and cons, most solutions work well for a specific scenario, asking for flexible but complex solutions.

References

Recommend Readings:

- J. Schiller: Mobile Communications, Addison-Wesley, 2003
- C. Beard, W. Stallings: Wireless Communication Networks and Systems, Pearson, 2016

Further Readings:

- W. Stallings: Data and Computer Communications, Pearson, 2014
- R. Freeman: Fundamentals of Telecommunications, Wiley, 2005